# From Noise to Diversity: Random Embedding Injection in LLM Reasoning

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## Abstract

Recent *soft prompt* methods, which prepend or insert trainable continuous embeddings into LLM inputs as virtual tokens, optimize these embeddings to enhance mathematical reasoning. However, the mechanism by which embedding injection itself affects model behavior, independent of optimization, remains unexplored. We investigate Random Soft Prompts (RSPs), continuous vectors sampled from an isotropic Gaussian fitted to the empirical mean and variance of the model's token embedding table and injected without any training. Empirically, RSPs increase early-stage token entropy while generation stabilizes thereafter, and combining RSPs with temperature sampling yields higher trajectory diversity as measured by Pass@$n$. Theoretically, we provide two complementary analyses: an energy-based framework that interprets RSP as an idealized stochastic perturbation of a Hopfield-style energy landscape with attention-mediated decay across layers and generation steps, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond temperature sampling. We further discuss the implications for GRPO training, where trajectory diversity is critical for learning signal quality.

## 1 Introduction

As large language models (LLMs) scale to billions of parameters, full fine-tuning becomes increasingly expensive. This has motivated parameter-efficient methods that adapt frozen models by introducing a small number of trainable parameters [Hu et al., 2022, Houlsby et al., 2019]. Among these, *soft prompts* prepend or insert learnable continuous embeddings into the input sequence, steering model behavior through the attention mechanism [Li and Liang, 2021, Lester et al., 2021]. This paradigm has been actively adopted for enhancing mathematical reasoning in LLMs [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2025]. Despite their diverse designs, these methods share a common structure: they inject continuous embeddings that lie outside the pretrained vocabulary and optimize them to improve downstream performance.

However, we find that optimization is not a necessary condition for these effects. We observe that injecting randomly initialized, untrained continuous embeddings can also shift the output distribution. For instance, applying a chat template to a base model not fine-tuned with such format causes a prompt-format mismatch that degrades reasoning capability; appending random embeddings to such mismatched inputs mitigates this degradation. This suggests that injecting out-of-vocabulary continuous embeddings itself, independent of any learned content, plays a non-trivial role in shaping model behavior. Existing studies primarily evaluate *soft prompt* methods through end-task accuracy, and the mechanisms by which embedding injection influences the output distribution remain insufficiently explored.

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In this work, we investigate the role of Random Soft Prompts (RSPs), continuous vectors sampled from an isotropic Gaussian fitted to the empirical mean and variance of the model's token embedding table and injected without any training. Empirically, RSPs increase early-stage token entropy while the output stabilizes in later generation stages, and combining RSPs with temperature sampling yields more diverse reasoning trajectories as measured by Pass@$n$ scaling. Theoretically, we analyze RSPs from two perspectives: an energy-based framework showing that RSP introduces controlled stochastic perturbations with natural decay through the attention mechanism, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond what temperature sampling can achieve. We further discuss the implications of these findings for Group Relative Policy Optimization (GRPO) [Shao et al., 2024], where trajectory diversity is essential for effective learning signals.

Our contributions are as follows:

- We empirically analyze how RSPs affect LLM generation, showing that RSPs increase early token entropy and produce more diverse reasoning trajectories when combined with temperature sampling.
- We provide theoretical analyses from two perspectives: an energy-based framework explaining why RSP can perturb the output distribution without degrading performance, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond the reach of temperature sampling.
- We discuss the implications of RSP-induced diversity for GRPO training and argue that RSP provides exploration gains complementary to existing temperature-based approaches.

# 2 Background

## 2.1 Soft Prompt Methods for LLM Reasoning

*Soft prompts* emerged as a parameter-efficient alternative to full fine-tuning. Prefix-Tuning [Li and Liang, 2021] introduced the paradigm by prepending trainable continuous vectors to every transformer layer, Prompt Tuning [Lester et al., 2021] simplified this to input-layer-only tuning and showed that it matches full fine-tuning at sufficient scale, and P-tuning [Liu et al., 2023] extended the approach with an LSTM-based prompt encoder to model inter-position dependencies.

More recently, *soft prompts* have been actively applied to LLM reasoning. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via entropy minimization, steering the model toward high-certainty states to activate latent reasoning capabilities. SoftCoT [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant model, leveraging the stronger representational capacity of continuous representations over discrete language. LTPO [Ye et al., 2026] optimizes latent thought embeddings per test instance through policy gradient with a confidence-based intrinsic reward. COCONUT [Hao et al., 2025] feeds the model's own hidden states back as input embeddings, enabling reasoning in a continuous latent space. MemGen [Zhang et al., 2025] demonstrates that embedding vectors can serve as experience memory by constructing latent token sequences that enrich the model's reasoning. Pause tokens [Goyal et al., 2024] insert learnable tokens into the input to provide additional computation steps.

These approaches all inject out-of-vocabulary continuous embeddings optimized for task-specific objectives. Because evaluation centers on end-task accuracy, the contribution of the injection itself and that of the optimization are not separated, and the mechanisms by which such out-of-vocabulary continuous embeddings shape model behavior remain a separate and underexplored question.

## 2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at various stages. During training, dropout [Srivastava et al., 2014] randomly deactivates hidden units to prevent overfitting, and NEFTune [Jain et al., 2024] adds noise to token embeddings during instruction fine-tuning, reporting improvements in instruction following. These methods use noise as a regularizer and do not apply perturbations at inference time.

At inference time, perturbation-based approaches primarily target robustness and safety. Randomized smoothing [Cohen et al., 2019] injects Gaussian noise into inputs to obtain certified robustness

Table 1: Accuracy (%) across model configurations and benchmarks. RSP reports the best result across injection positions (Prefix, Infix, Suffix); full per-position breakdown is in Appendix A. Values in parentheses indicate the difference from the baseline.

| | **MATH-500** | | **GSM8K** | | **AIME24** | |
|---|---|---|---|---|---|---|
| **Model** | Baseline | RSP | Baseline | RSP | Baseline | RSP |
| *Instruct Models* | | | | | | |
| LLaMA-3.1-8B-Inst | 52.20 | 49.40 (−2.8) | 85.75 | 86.73 (+1.0) | 6.67 | 10.00 (+3.3) |
| Qwen2.5-Math-7B-Inst | 83.20 | 84.20 (+1.0) | 95.45 | 95.68 (+0.2) | 13.33 | 16.67 (+3.3) |
| Qwen2.5-Math-1.5B-Inst | 73.00 | 75.20 (+2.2) | 85.29 | 85.75 (+0.5) | 10.00 | 20.00 (+10.0) |
| *Base Models (Raw Text)* | | | | | | |
| Qwen2.5-Math-7B | 72.20 | 71.60 (−0.6) | 84.15 | 84.31 (+0.2) | 6.67 | 16.67 (+10.0) |
| Qwen2.5-Math-1.5B | 61.40 | 64.40 (+3.0) | 77.86 | 74.30 (−3.6) | 16.67 | 10.00 (−6.7) |
| *Base Models + ChatML (Format Mismatch)* | | | | | | |
| Qwen2.5-Math-7B | 52.20 | 70.40 (+18.2) | 58.30 | 75.97 (+17.7) | 23.33 | 23.33 (+0.0) |
| Qwen2.5-Math-1.5B | 34.40 | 63.40 (+29.0) | 37.45 | 67.02 (+29.6) | 13.33 | 20.00 (+6.7) |

guarantees. In the LLM setting, SmoothLLM [Robey et al., 2023] perturbs input prompts at the character level to defend against adversarial jailbreaking. These methods require multiple noisy forward passes with output aggregation, and target prediction stability rather than output diversity.

In reinforcement learning, NoisyNet [Fortunato et al., 2018] injects learnable noise into network weights to improve exploration, demonstrating that the location and structure of noise injection affect exploration quality. This motivation is closest to RSPs, which similarly target diversity, but differ in injection target (input embeddings vs. network weights) and require no learned noise parameters.

RSPs differ from these approaches in the following respects. **(1)** RSPs operate solely at inference without any training, whereas dropout and NEFTune inject noise during training, modifying the resulting model parameters accordingly. **(2)** RSPs preserve the original input and append fresh embedding vectors in a single forward pass, whereas robustness methods corrupt the input and aggregate over multiple noisy passes. **(3)** RSPs use purely random vectors sampled from the pretrained embedding statistics, whereas NoisyNet learns noise parameters through optimization.

# 3 Experimental Analysis

## 3.1 Random Soft Prompt: Definition and Setup

Let $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ denote the token embedding matrix of a pretrained LLM, where $V$ is the vocabulary size and $d$ is the hidden dimension. We define an RSP $\mathbf{H} = [h_1, \ldots, h_L] \in \mathbb{R}^{L \times d}$ as a sequence of $L$ continuous vectors, where each vector is independently sampled from:

$$h_j \sim \mathcal{N}(\mu_E,\ \sigma_E^2 \mathbf{I}), \quad j = 1, \ldots, L, \tag{1}$$

with $\mu_E = \frac{1}{Vd} \sum_{i,k} [\mathbf{W}_E]_{i,k}$ and $\sigma_E = \mathrm{std}(\mathbf{W}_E)$. Let $\mathbf{X} = [x_1, \ldots, x_T] \in \mathbb{R}^{T \times d}$ denote the embedding sequence of the original input. The RSP-augmented input $\tilde{\mathbf{X}}$ is constructed by concatenating $\mathbf{H}$ with $\mathbf{X}$ at one of three positions without modifying the original prompt text: *prefix* ($[\mathbf{H}; \mathbf{X}]$), *infix* ($\mathbf{H}$ inserted within $\mathbf{X}$), or *suffix* ($[\mathbf{X}; \mathbf{H}]$), following the injection strategies adopted in prior work [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2025]. Sampling a new $\mathbf{H}$ for each batch ensures that the results are not dependent on any realization of the random embeddings.

We evaluate on three mathematical reasoning benchmarks: MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across seven configurations including instruct models, base models, and base models with mismatched chat templates. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity. Only the final answer is evaluated through a fallback-based extraction pipeline. Full experimental details are provided in Appendix A.

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

| Model | Benchmark | Baseline | RSP | TTSV | SoftCoT | LTPO |
|---|---|---|---|---|---|---|
| Qwen2.5-Math-7B-Instruct | MATH-500 | 83.20 | **84.20** | 83.80 | 82.80 | 83.00 |
| | GSM8K | 95.45 | **95.68** | 95.30 | 95.00 | 94.84 |
| | AIME24 | 13.33 | 16.67 | **23.30** | 16.67 | 13.33 |
| Qwen2.5-Math-1.5B-Instruct | MATH-500 | 73.00 | 75.20 | 75.20 | **76.00** | 72.40 |
| | GSM8K | 85.29 | **85.75** | 85.70 | 84.46 | 84.53 |
| | AIME24 | 10.00 | **20.00** | 6.70 | 6.67 | 10.00 |

### 3.2 Does RSP Degrade Reasoning Performance?

We first examine how injecting random embeddings affects the model's reasoning performance. Table 1 reports the accuracy for each model configuration, using the best RSP result across injection positions (Prefix, Infix, Suffix) and $L \in \{10, 15, 20\}$ for each benchmark.

For instruct models and base models with their native prompt formats, RSP injection maintains or slightly shifts baseline performance within a few percentage points. Most strikingly, under prompt-format mismatch — where base models receive ChatML templates they were not trained on — RSP yields pronounced improvements of up to $+29\%$p. These gains are difficult to attribute to simple perturbation effects, as noise that merely raises model uncertainty would further degrade performance.

We compare RSPs with prior soft prompt methods (TTSV, SoftCoT, LTPO) that optimize their embeddings for distinct objectives, under a unified evaluation protocol reported in Table 2. All methods are re-evaluated with a shared answer extraction pipeline that applies fallback strategies, enabling a format-agnostic comparison despite each method's distinct prompt format and grading scheme. The results place RSPs on par with the optimized methods without requiring any training.

### 3.3 How Does RSP Affect the Output Distribution?

We analyze how RSP injection changes the model's output distribution during generation on Qwen2.5-Math-1.5B-Instruct with MATH-500, measuring per-token entropy, top-1 probability, and varentropy across the generation process. Figure 1 and Table 3 compare these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods.

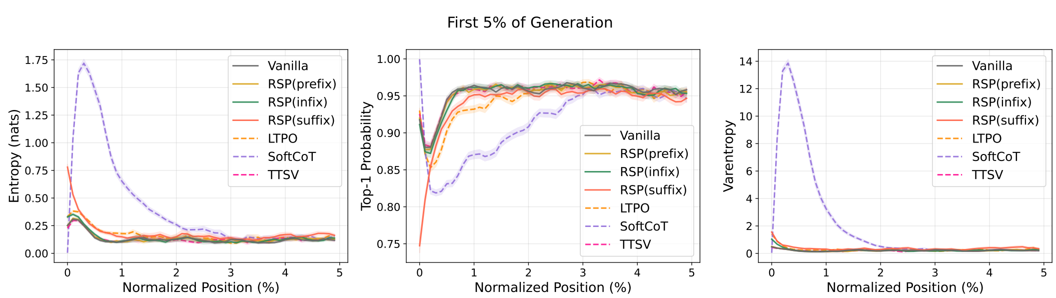


Figure 1: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates $\pm 1$ SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods (LTPO, SoftCoT, TTSV).

Table 3: Mean entropy, top-1 probability, and varentropy for the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500).

| Metric | Baseline | Prefix | Infix | Suffix | LTPO | SoftCoT | TTSV |
|---|---|---|---|---|---|---|---|
| Entropy | 0.1332 | 0.1366 | 0.1402 | 0.1941 | 0.1662 | 0.4218 | 0.1359 |
| Top-1 Prob | 0.9535 | 0.9523 | 0.9525 | 0.9373 | 0.9437 | 0.9156 | 0.9534 |
| Varentropy | 0.2181 | 0.2207 | 0.2463 | 0.4010 | 0.2953 | 2.2234 | 0.2174 |

These results show that RSP modifies the early-stage output distribution across all injection positions: entropy increases (the token distribution flattens), top-1 probability decreases (commitment to a dominant token weakens), and varentropy increases (indicating multimodal distributions [Ahmed et al., 2026]). High-entropy positions are known to act as critical forking points that determine reasoning trajectories [Wang et al., 2025], so this shift moves the model toward a state where multiple reasoning paths can coexist. The change is localized to the initial reasoning stage; in the middle and final portions of generation, all three metrics converge to baseline levels.

Although existing soft prompt methods are optimized for different objectives, they share a common pattern of increasing early-stage entropy. Notably, TTSV, which optimizes soft prompts using a reward that reduces the mean entropy of reasoning trajectories, still exhibits early entropy comparable to the baseline. This suggests that elevating early-stage entropy is an inherent effect of injecting non-pretrained continuous embeddings, independent of the optimization objective.

### 3.4 Does RSP Induce Trajectory Diversity?

Section 3.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

**Token-level: distribution beyond temperature.** Under autoregressive decoding, any trajectory-level divergence between RSP and the baseline must originate from the first generation step, a critical forking point for reasoning trajectories [Wang et al., 2025]. Using the baseline at $\tau$=1.0 as reference, we compare RSP at $\tau$=1.0 against the baseline at $\tau$=2.0 (the high-temperature ceiling of what flattening alone can achieve) on three complementary metrics: Spearman rank correlation $\rho$ to detect rank reorganization, the probability mass placed outside the baseline's top-10 to measure support expansion, and Jensen–Shannon divergence to quantify the overall magnitude of distributional change (formal definitions in Appendix C).

Table 4: First-token distribution metrics measured against the baseline at $\tau$=1.0, comparing the baseline at $\tau$=2.0 with RSP at $\tau$=1.0.

| Metric | $\tau$=2.0 | RSP |
|---|---|---|
| Spearman $\rho$ | 1.0 | **0.651** |
| Mass outside top-10 | 5.15% | **27.64%** |
| JS divergence | 0.086 | **0.231** |

Table 4 reports the contrast. Even at $\tau$=2.0, only 5.15% of probability mass falls outside the baseline's top-10; RSP at $\tau$=1.0 reaches 27.64%. The Spearman rank correlation tells the story more starkly: because temperature scaling $p_i^{(\tau)} \propto p_i^{1/\tau}$ is a monotone transform that preserves token ranking ($p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$) for any $\tau > 0$, $\rho$ relative to the baseline is exactly $1$ as a mathematical identity. RSP, by contrast, reranks the token distribution itself, reducing $\rho$ to $0.651$—an effect temperature cannot produce at any $\tau$. The Jensen–Shannon divergence ($0.231$) quantifies the overall magnitude of this distributional change. Since $\rho < 1$ rules out rank-preserving flattening as the sole cause, the high divergence reflects a structural reshape of the distribution rather than a uniform flattening of the same ranking.

**Trajectory-level: semantic diversity.** Token-level reranking raises a second question: do first-token perturbations produce semantically distinct reasoning paths, or do independently perturbed trajectories converge to the same solution strategy? We randomly sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under two conditions, Baseline and RSP, with temperature $\tau = 1.0$. The trajectories are embedded via BGE-M3 [Chen et al., 2024] and compared on three metrics: mean pairwise cosine distance within each problem, inter-cluster distance between DBSCAN [Ester et al., 1996] centroids, and intra-cluster distance. Table 5 shows that RSP substantially increases the inter-cluster distance while leaving the intra-cluster distance unchanged: clusters become more separated while reasoning within each cluster remains consistent. RSP also wins on pairwise distance in 56 of 64 problems (87.5%).

Table 5: Semantic diversity metrics averaged across the 64 problems.

| Metric | Baseline | RSP |
|---|---|---|
| Pairwise cos. dist. | 0.0612 | **0.0879** |
| Inter-cluster dist. | 0.0183 | **0.0982** |
| Intra-cluster dist. | 0.0526 | 0.0528 |

Figure 2(a) illustrates the qualitative difference for the representative problem: the RSP samples (blue) cover the region occupied by baseline samples (red) while extending into clusters that the

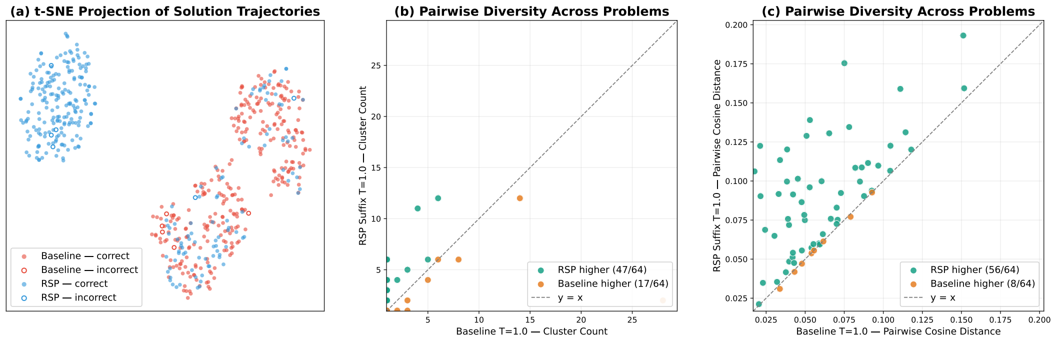


Figure 2: Semantic diversity analysis on 64 randomly sampled MATH-500 problems with 300 trajectories per condition (Qwen2.5-Math-1.5B-Instruct, $\tau = 1.0$, BGE-M3 embeddings). (a) t-SNE projection for one representative problem (`test/prealgebra/951`): baseline (red) concentrates around one region while RSP suffix (blue) spreads across distinct clusters; filled markers denote correct answers and open markers denote incorrect. (b) Per-problem mean pairwise cosine distance; points above the diagonal indicate RSP produces more diverse trajectories than baseline. (c) Per-problem DBSCAN cluster count; RSP yields more distinct clusters than baseline on most problems.

baseline does not reach; the expanded regions also contain correct solutions (filled markers), showing that RSP's exploration is not limited to incorrect alternatives but reaches new reasoning paths that still yield the correct answer. Figures 2(b) and 2(c) show that this pattern holds across the 64 problems: on most problems RSP produces more clusters and larger pairwise cosine distance than baseline. The combination of large inter-cluster increase and unchanged intra-cluster distance mirrors the token-level finding: RSP alters which cluster a trajectory enters while preserving the local continuation within each cluster. Additional per-problem t-SNE plots are provided in Figure 6 (Appendix B).

**Outcome-level: Pass@$n$ scaling.** Finally, we evaluate whether the token- and trajectory-level diversity translates into task-level outcome diversity through Pass@$n$, the probability that at least one of $n$ sampled solutions is correct. We compare four conditions with $n = 16$ samples per problem on MATH-500 and AIME24: *(i) Baseline,* $\tau = 1.0$: temperature sampling only; *(ii) RSP (fixed seed),* $\tau = 1.0$: a single RSP shared across all samples; *(iii) RSP (indep. seed), greedy*: a different RSP per sample without temperature; and *(iv) RSP (indep. seed),* $\tau = 1.0$: a different RSP per sample combined with temperature sampling.

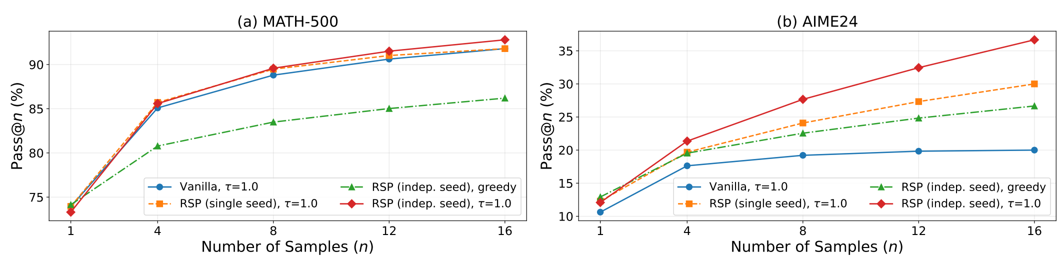


Figure 3: Pass@$n$ scaling on MATH-500 (left) and AIME24 (right) with Qwen2.5-Math-1.5B-Instruct, $n = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

Condition *(iv)* achieves the highest Pass@$n$ on both benchmarks, reaching 92.80% on MATH-500 and 36.67% on AIME24 at $n = 16$. Condition *(ii)* shows no improvement over the baseline, confirming that the diversity gain originates from varying the random embeddings across samples, not from a single fixed perturbation. Pass@1 remains comparable to the baseline on MATH-500, and on the more challenging AIME24, *(iv)* achieves higher Pass@1 than the baseline. Independent RSPs combined with temperature sampling produce the most diverse reasoning trajectories while maintaining or improving single-sample accuracy.

**Synthesis across the three levels.** The three analyses converge on a consistent picture. At the token level, RSP induces support expansion concentrated at the first token and decaying monotonically across steps. At the trajectory level, this localized perturbation translates into distinct reasoning strategies (large inter-cluster separation) while preserving local coherence within each strategy (unchanged intra-cluster distance). At the outcome level, the resulting trajectories cover more correct solutions without sacrificing single-sample accuracy.

# 4 Theoretical Analysis: Why RSP Works

Our theoretical goal is narrower than a theorem about the full nonlinear transformer. We want a mechanism that explains the three empirical regularities from Section 3: random continuous prompts can alter the model's earliest reasoning decisions even though they are not optimized for the task; their influence is strongest at the beginning of decoding and weakens later; and this early perturbation can increase trajectory diversity without permanently destabilizing generation. The common control variable throughout the section is $w_{r,i}^{(l)}$, the attention mass assigned to the random tokens. The argument therefore has three parts. First, we characterize what kind of perturbation can open reasoning branches without inserting a hand-crafted prior, and explain why centered isotropic randomness is the relevant choice. Second, we show that attention injects the perturbation through $w_{r,i}^{(l)}$ and simultaneously places an annealing pressure on it as real-token evidence grows. Third, we show why this pre-softmax perturbation can change token rankings in a way that temperature scaling cannot.

## 4.1 Why Random Rather Than a Fixed Token?

A generic exploration prompt for reasoning should satisfy three structural requirements. It should not systematically pre-commit the model to one semantic direction before the input-dependent computation begins. It should vary across trials, otherwise repeated decoding would trace the same perturbation every time. And because the branch-sensitive direction is unknown in advance and depends on the prompt, the decoding state, and the token comparison, the perturbation should not privilege some directions while leaving others almost untouched. Optimized soft prompts deliberately encode a task prior [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2025]; our question is the complementary one: what perturbation should one use when no such prior is assumed?

A fixed token or a fixed continuous prompt fails this requirement set immediately. It may steer one trajectory, but if the same vector is reused across runs then there is no across-trial exploration at all. Resampling discrete vocabulary tokens restores variability, yet each draw still comes with its own lexical direction and there is no reason for the resulting perturbations to be centered or isotropic. We formalize the choice we make instead.

**Definition 1** (Random Soft Prompt). *Let $\mathbf{E} \in \mathbb{R}^{|\mathcal{V}| \times d}$ denote the pretrained input-embedding table, and define the empirical entrywise mean and variance*

$$\bar{\mu}_E := \frac{1}{|\mathcal{V}|d} \sum_{v \in \mathcal{V}} \sum_{m=1}^{d} E_{v,m}, \qquad \sigma_E^2 := \frac{1}{|\mathcal{V}|d} \sum_{v \in \mathcal{V}} \sum_{m=1}^{d} (E_{v,m} - \bar{\mu}_E)^2, \qquad \mu_E := \bar{\mu}_E \mathbf{1}_d.$$

*In the implementation studied here, an* RSP *of length $L$ is a sequence $\mathbf{H} = (h_1, \ldots, h_L)$ with*

$$h_l = \mu_E + \sigma_E \xi_l, \qquad \xi_l \overset{iid}{\sim} \mathcal{N}(0, \mathbf{I}_d).$$

*Equivalently, the centered prompt $\bar{h}_l := h_l - \mu_E$ satisfies $\bar{h}_l \sim \mathcal{N}(0, \sigma_E^2 \mathbf{I}_d)$. Whenever we later analyze repeated-sampling protocols such as Pass@$N$, we additionally assume the draws used across those compared trials are independent.*

The proofs below use only the centered law of $\bar{h}_l$, not the particular form of the centering vector $\mu_E$. The scalar fit $\mu_E = \bar{\mu}_E \mathbf{1}_d$ therefore matches the implementation without changing the subsequent arguments. The structural ingredients are centeredness, isotropic covariance, and fresh resampling whenever multiple RSP trials are compared. The two embedding-derived statistics in Definition 1 play distinct roles. The scalar $\sigma_E^2$ borrows the *scale* of the pretrained embedding distribution so that the perturbation operates in the magnitude regime the model expects. The covariance $\sigma_E^2 \mathbf{I}_d$, in contrast, deliberately discards the directional structure of the embedding distribution; Theorem 1 below shows that this is the unique covariance that maximizes the worst-case directional variance under a fixed trace budget. The downstream attention mechanism then maps this prompt-space distribution to value space via the value projection $\mathbf{W}_V^{(l)}$; the corresponding variance facts are stated explicitly below.

**Lemma 1** (Semantic neutrality)**.** *Let* $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ *and* $\hat{h} := \bar{h}/\|\bar{h}\|$*. For any fixed unit vectors* $u_1, \ldots, u_M \in \mathbb{S}^{d-1}$ *and any* $\delta \in (0, 1)$*,*

$$\Pr\left[\max_{r \in [M]} |\hat{h}^\top u_r| \leq \sqrt{\frac{2\log(2M/\delta)}{d-1}}\right] \geq 1 - \delta.$$

*In contrast, for any nonzero token embedding* $e_s$*, choosing the anchor direction* $u = \hat{e}_s := e_s/\|e_s\|$ *yields* $|\hat{e}_s^\top u| = 1$*.*

Lemma 1 gives the precise sense in which the perturbation is neutral. The claim is not that a random vector is literally free of every incidental semantic correlation. The relevant and weaker statement is that it does not systematically align with any pre-selected finite family of semantic alternatives. Centering then removes deterministic first-order drift: under the local surrogate below, the expected perturbation of any logit gap is zero when $\mathbb{E}[\bar{h}] = 0$.

To formalize the exploration part, let $\bar{h}$ be the centered prompt vector and consider the local first-order surrogate

$$z_i(\bar{h}) \approx z_i + a_i^\top \bar{h}, \qquad a_i := \nabla_{\bar{h}} z_i(0),$$

valid for sufficiently small $\|\bar{h}\|$ (Appendix E); we use this surrogate throughout the section to identify the direction of perturbation rather than to predict its exact magnitude. For a token pair $(i, j)$, the perturbed logit gap becomes

$$\Delta_{ij}(\bar{h}) \approx \Delta_{ij} + b_{ij}^\top \bar{h}, \qquad b_{ij} := a_i - a_j.$$

The relevant question is therefore not "does the perturbation have variance?" but "how much variance does it retain along the unknown direction $b_{ij}$ that matters for this branching decision?" The theorem below answers that question at the level of second moments.

**Theorem 1** (Direction-agnostic coverage)**.** *Let $D$ be any zero-mean perturbation distribution on* $\mathbb{R}^d$ *with covariance* $\Sigma_D$ *and variance budget* $\mathrm{tr}(\Sigma_D) \leq \rho^2$*. Define the worst directional variance*

$$\mathcal{E}(D) := \min_{\|b\|=1} \mathrm{Var}_{h \sim D}(b^\top h).$$

*Then*

$$\mathcal{E}(D) = \lambda_{\min}(\Sigma_D) \leq \frac{\rho^2}{d},$$

*with equality if and only if* $\Sigma_D = (\rho^2/d)\mathbf{I}_d$*. Hence isotropic covariance is the unique way to equalize first-order perturbation variance across all directions while maximizing the least-covered one.*

Theorem 1 is not meant to cast prompting as an adversarial game; its role is simpler. When the branch-sensitive direction is unknown at sample time, isotropy is the unique second-order description that neither favors nor neglects any direction. A fixed prompt has $\Sigma_D = 0$ and therefore creates no across-trial exploration at all. An anisotropic perturbation leaves at least one direction less explored than the isotropic optimum. The Gaussian choice in RSP is the simplest centered, rotation-symmetric family that realizes this covariance while matching the scale of pretrained embeddings. This is the sense in which RSP is not just "some noise": it is a resampled, centered, direction-agnostic perturbation.

A natural alternative one might consider is to draw the perturbation from a distribution that *matches* the empirical geometry of the pretrained embedding table $\mathbf{W}_E$, rather than discarding it. Theorem 1 dismisses this alternative as soon as the embedding geometry is known to be anisotropic.

**Corollary 1** (Embedding-matched noise is suboptimal in worst-direction coverage)**.** *Let* $\Sigma_E$ *be the empirical covariance of the rows of* $\mathbf{W}_E$*, and let* $\rho^2 := \mathrm{tr}(\Sigma_E)$*. Consider two zero-mean perturbations on* $\mathbb{R}^d$ *with the same trace budget* $\rho^2$*: the isotropic RSP law* $D_{\mathrm{iso}}$ *with covariance* $(\rho^2/d)\mathbf{I}_d$*, and the embedding-matched law* $D_{\mathrm{emb}}$ *with covariance* $\Sigma_E$*. Then*

$$\mathcal{E}(D_{\mathrm{iso}}) \;=\; \frac{\rho^2}{d} \;\geq\; \lambda_{\min}(\Sigma_E) \;=\; \mathcal{E}(D_{\mathrm{emb}}),$$

*with strict inequality whenever* $\Sigma_E$ *is non-isotropic.*

The corollary is one application of Theorem 1 and requires no separate proof. Its empirical bite comes from the fact that pretrained LLM token-embedding matrices are documented to be highly anisotropic, concentrating in a low-dimensional "cone" inside $\mathbb{R}^d$ [Mu and Viswanath, 2018, Ethayarajh, 2019]. In that regime $\lambda_{\min}(\Sigma_E)$ is orders of magnitude smaller than $\rho^2/d$, so the worst-covered direction under embedding-matched noise is nearly unexplored—precisely the directions a generic exploration prompt should not abandon. Discarding the directional geometry of $\mathbf{W}_E$ while keeping its scale is therefore not a simplification of an embedding-matched alternative but a quantitatively better choice for direction-agnostic exploration.

Lemma 1 and Theorem 1 therefore play complementary roles. The lemma rules out semantic pre-commitment to any fixed anchor set, while the theorem rules out directional under-coverage in the local branch-sensitive geometry. A perturbation that satisfies only the first condition would be neutral but too weak in some directions; a perturbation that satisfies only the second would still inject a hand-chosen semantic prior. RSP is useful because it satisfies both at once.

The two results together also pin down the distribution itself, not merely its second moments. Among all centered distributions with isotropic covariance, the Gaussian uniquely maximizes differential entropy under the variance budget, formalizing the sense in which RSP injects no additional structural prior beyond its first two moments.

**Theorem 2** (Characterization of generic-exploration distributions)**.** *Fix a variance budget $\rho^2 > 0$ and a dimension d. Among all probability distributions on $\mathbb{R}^d$ that are **(C1)** centered ($\mathbb{E}[h] = 0$), **(C2)** have variance budget $\operatorname{tr}(\Sigma_D) \le \rho^2$, and **(C3)** achieve the minimax bound $\mathcal{E}(D) = \rho^2/d$ of Theorem 1, the centered isotropic Gaussian $\mathcal{N}(0, (\rho^2/d)\,\mathbf{I}_d)$ is the unique distribution that additionally maximizes differential entropy.*

Theorem 2 is the formal statement of "why exactly this distribution and not another random one." (C1) forces centeredness, (C3) plus the equality case of Theorem 1 forces isotropic covariance, and the maximum-entropy condition selects Gaussianity [Cover and Thomas, 2006]. The same desiderata rule out concrete alternatives at the second-moment level alone: PCA-aligned noise on the top-$k$ embedding eigenvectors yields $\Sigma_D$ supported on a $k$-dimensional subspace and fails (C3); random discrete vocabulary tokens yield $\Sigma_D$ supported on the embedding manifold (measure zero in $\mathbb{R}^d$) and again fail (C3); a reused soft prompt has $\Sigma_D = 0$ and fails (C3) trivially. RSP is the unique entry in this list that achieves equality in (C3) and the maximum-entropy condition.

### 4.2 How Attention Turns a Neutral Perturbation into Controlled Exploration

The previous subsection explains why centered isotropic randomness is the right kind of perturbation. The next question is temporal: why is the perturbation strong enough to matter at the beginning of decoding but weak enough not to derail the rest of the reasoning chain? The formal answer uses three ingredients. First, in an idealized Hopfield-style view of attention, each transformer block acts like one step of an iterative attractor process; in that picture the unperturbed path is strongly constrained by its initialization, which identifies a structural source of exploration deficit. Second, the attention output admits an exact decomposition into a real-token update and a random-token contribution. Third, the same attention mass that injects the random contribution is structurally forced downward as real-token evidence accumulates, and the perturbation propagates through the feed-forward sublayer with controlled amplification.

**Iterative attractor view.** Following the Modern Hopfield correspondence [Ramsauer et al., 2021], one attention update with key matrix $\mathbf{K}$ and value matrix $\mathbf{V}$ is equivalent to a gradient step on the energy

$$E_{\mathbf{K}}(\sigma) = -\beta^{-1} \log \sum_j \exp(\beta\, \mathbf{k}_j^\top \sigma) + \tfrac{1}{2}\|\sigma\|^2$$

when $\mathbf{K} = \mathbf{V}$. Under three idealized assumptions *(A1)* $\mathbf{K} = \mathbf{V}$*, (A2) weight-tied attention, (A3) $L_E$-smoothness of $E_{\mathbf{K}}$ with $L_E < 2$,* Appendix E.1 shows that the unperturbed iterates produce a strictly monotone descent in $E_{\mathbf{K}}$ and the sublevel set $\{\sigma : E_{\mathbf{K}}(\sigma) \le E_{\mathbf{K}}(\sigma^{(0)})\}$ is forward-invariant; under the additional separation condition $\operatorname{sep}(C(\sigma^{(0)})) > 0$ together with the per-step gradient bound of Corollary 3, the trajectory also remains in the connected component containing $\sigma^{(0)}$.

(A1) and (A2) are not literally satisfied by real transformers, so the Hopfield view here is an idealized structural picture rather than a literal statement about every transformer block. Its role is to identify why a deterministic forward pass can be exploration-poor: under A1–A3 it stays inside a fixed sublevel set, and under the extra separation/small-step condition it also remains in the connected component selected by initialization. We use this picture to motivate why injecting an isotropic stochastic perturbation at every block is the relevant intervention, not to claim formal escape rates for real networks.

Throughout the rest of the section we work with the actual transformer attention. The random values are obtained by applying the value projection to the centered prompt vectors, so

$$\mathbf{v}_{r_j}^{(l)} = \mathbf{W}_V^{(l)} \bar{h}_j, \qquad \mathrm{Cov}(\mathbf{v}_{r_j}^{(l)}) = \sigma_E^2 \mathbf{W}_V^{(l)} \mathbf{W}_V^{(l)\top}.$$

Proposition 5 in Appendix D states the resulting directional-variance bounds explicitly. In particular, if $\mathbf{W}_V^{(l)}$ is full rank then the projected perturbation remains non-degenerate in every value-space direction, and if $\mathbf{W}_V^{(l)}$ is well-conditioned then the prompt-space direction-agnostic coverage carries to value space up to layer-dependent constants.

**Exact decomposition.** When $k$ random vectors are inserted as additional keys and values, the attention update can be written as

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)})\, \tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)}, \tag{2}$$

where $\tilde{\mathbf{x}}_i^{(l+1)}$ is the update produced by the real tokens alone, $\boldsymbol{\eta}_i^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}$ is the contribution of the random values, and

$$w_{r,i}^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)}$$

is the total attention mass assigned to the random tokens. Appendix E.2 proves the decomposition and the deterministic bound

$$\|\boldsymbol{\eta}_i^{(l)}\| \le w_{r,i}^{(l)} \max_{j\in[k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

This is the structural core of the mechanism: the same scalar $w_{r,i}^{(l)}$ controls how much of the original update is retained and how large the random contribution can be. If $w_{r,i}^{(l)}$ is large, the perturbation can materially change the update; if $w_{r,i}^{(l)}$ is small, the computation collapses back toward the real-token update.

**Natural annealing through attention.** The second structural fact is that $w_{r,i}^{(l)}$ is not arbitrary. It satisfies

$$w_{r,i}^{(l)} \le \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}, \tag{3}$$

where $\Delta_i^{(l)} := \min_{j\in[n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j'\in[k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ is the real-vs-random logit gap and $n$ is the number of real tokens. This bound yields a clean *sequence-wise dilution effect*: for a fixed gap, every newly generated real token tightens the maximum random-token attention compatible with that state. A similar *layer-wise* decay is expected whenever deeper layers enlarge the real-vs-random gap $\Delta_i^{(l)}$. We intentionally treat that layer-wise statement as a model-side hypothesis rather than as a universal theorem, because it depends on how the model sharpens real-token retrieval with depth. The resulting prediction is simple: the perturbation should be strongest exactly when the model has the least context and many continuations remain open, and it should weaken as evidence accumulates.

**Theorem 3** (Sequence-wise annealing under a fixed logit gap)**.** *Fix $k$ and $\Delta_i^{(l)}$. The upper bound in Eq.* (3) *is strictly decreasing in the number $n$ of real tokens. Consequently, cache growth alone imposes a monotone decrease on the largest random-token attention compatible with that logit gap.*

Theorem 3 proves a structural dilution effect, not monotone decay of the realized $w_{r,i}^{(l)}$ on every decoding path, because $\Delta_i^{(l)}$ also evolves with state. The layer-wise half is likewise model-specific and must be checked empirically.

**Per-block basin escape under directional coverage.** The decomposition together with the Gaussian sampling of random values yields a directional escape probability at the post-attention level of every block. Suppose the unperturbed post-attention state at block $l$ lies in a basin $B$ of $E_{\mathbf{K}}$ whose boundary along a unit direction $u$ is at distance $r_u^{(l)} > 0$. Conditional on the attention weights, $u^\top \boldsymbol{\eta}_i^{(l)}$ is exactly Gaussian with variance $\sigma_E^2 \|\alpha_R^{(l)}\|^2$, where $\alpha_R^{(l)}$ collects the random-token weights. Cauchy–Schwarz gives $\|\alpha_R^{(l)}\|^2 \geq (w_{r,i}^{(l)})^2/k$, so the variance is positive whenever $w_{r,i}^{(l)} > 0$, and Appendix E.5 shows

$$\Pr\Big(\text{post-attention state crosses } \partial B \text{ along } u\Big) \;\geq\; \Phi\left(-\frac{(r_u^{(l)} + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|)\sqrt{k}}{\sigma_E \, w_{r,i}^{(l)}}\right) > 0, \quad (4)$$

simultaneously for every $u$ with positive boundary distance. This is the formal sense in which an isotropic perturbation enables direction-agnostic escape at every block where $w_{r,i}^{(l)}$ is bounded away from zero: the same mechanism that achieves minimax coverage in the logit space (Theorem 1) ensures positive variance along every barrier-orthogonal direction in the hidden-state space.

The bound (4) is per-block and not tight. We use it to certify that escape probability is bounded below by a positive function of $w_{r,i}^{(l)}$ and the local barrier geometry. When $w_{r,i}^{(l)}$ is large at early blocks, escape probability is non-vanishing in every direction; when $w_{r,i}^{(l)}$ has decayed at later blocks, the dynamics approach deterministic descent within whichever basin the trajectory has reached, so the trajectory stabilizes. This combination of early escape and late stabilization is the formal counterpart of the empirical pattern reported in Section 3.4: RSP-induced trajectories diverge from the baseline early and then settle into different reasoning clusters.

**Feed-forward propagation under local Lipschitz control.** The decomposition above governs the attention sublayer. A real transformer block applies a feed-forward sublayer $F^{(l)}$ after attention with residual connections, so the cross-block perturbation depends on how $F^{(l)}$ propagates $\boldsymbol{\eta}_i^{(l)}$. If $F^{(l)}$ is $L^{(l)}$-Lipschitz on a bounded set containing both the perturbed and unperturbed post-attention states, then Appendix E.4 shows

$$\|\mathbf{x}_i^{(l+1,\text{RSP})} - \mathbf{x}_i^{(l+1,\text{base})}\| \leq (1 + L^{(l)}) \left(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)} \|\tilde{\mathbf{x}}_i^{(l+1)}\|\right),$$

which together with $\|\boldsymbol{\eta}_i^{(l)}\| \leq w_{r,i}^{(l)} \max_j \|\mathbf{v}_{r_j}^{(l)}\|$ shows that the per-block perturbation amplitude scales linearly with $w_{r,i}^{(l)}$. The bounded-set assumption is plausible in practice because LayerNorm constrains the post-attention norm, so the perturbed and unperturbed states traversed during a single forward pass lie in a common bounded region. Iterating across the fixed finite stack yields the explicit cumulative bound of Corollary 7. Under standard architectural assumptions this bound is finite for a fixed model depth, and the later-block injections themselves shrink as $w_{r,i}^{(l)}$ decreases. Stronger uniform-in-depth non-expansion would require additional contraction assumptions, which we do not invoke here.

**Empirical verification.** We test exactly these model-dependent steps on Qwen2.5-Math-7B under suffix injection. If $w_{r,i}^{(l)}$ really acts as an annealed exploration coefficient, then the realized attention to RSP tokens should decay across reasoning positions and layers, whereas attention to question tokens should not exhibit the same monotone pattern because they remain task-relevant throughout decoding. Figure 4 shows precisely this contrast. The figure therefore serves a specific logical role: it validates the model-side sharpening behavior that Eq. (3) by itself does not prove.

### 4.3 Output Distribution Shift and Exploration

The practical consequence of the previous mechanism is branch opening. To connect that mechanism to observable decoding behavior, we now return to the output distribution. The relevant question is not whether RSP merely flattens probabilities, but whether it can make the decoder consider tokens that temperature scaling can never promote into contention.

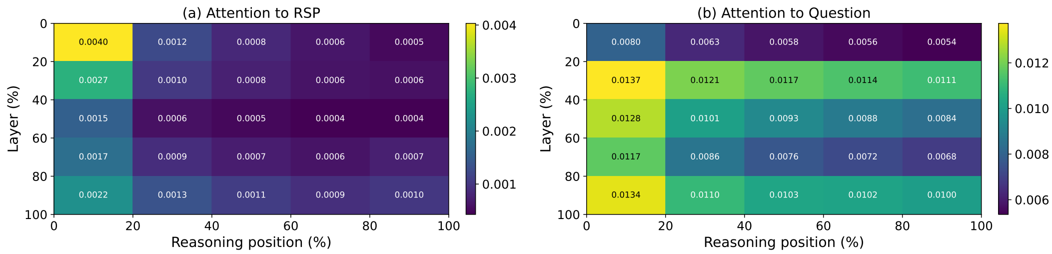


Figure 4: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). The layer and reasoning-position axes are each partitioned into five quantile bins; every cell reports the 500-sample mean per-token attention mass directed to **(a)** the RSP embeddings and **(b)** the question tokens within that region. The horizontal axis spans reasoning position from early (left) to late (right) generation steps, and the vertical axis spans layer depth from shallow (top) to deep (bottom). Attention to the RSP embeddings in (a) decays along both axes, while attention to the question tokens in (b) remains comparatively uniform.

**Why temperature is insufficient.** Temperature scaling transforms probabilities as $p_i^{(\tau)} \propto p_i^{1/\tau}$ and preserves token ranking, i.e., $p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$. It can soften or sharpen confidence, but it cannot create a new ordering among candidate tokens at a fixed decoding state.

**Why RSP changes rankings.** RSP acts before softmax, through the hidden state. At a fixed decoding state, temperature rescales existing logit gaps, whereas RSP perturbs the gaps themselves. Using the same first-order local surrogate as above, define

$$z_i^{\mathrm{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}.$$

The next two propositions are statements about these surrogate logits, not the exact transformer logits; this scope is sufficient for the comparison to temperature because temperature is rank-preserving regardless of approximation.

The next two propositions use only one additional support property of the centered prompt law: every nonempty open subset of $\mathbb{R}^d$ receives positive probability. The Gaussian instantiation of Definition 1 satisfies this automatically, but the proofs themselves do not require Gaussian tails once this support condition is in place.

**Proposition 1** (Stochastic ranking under the local affine surrogate)**.** *If there exists a token pair $(i, j)$ such that $a_i \neq a_j$, and if the centered prompt law assigns positive probability to every nonempty open subset of $\mathbb{R}^d$, then*

$$0 < \Pr_{\bar{h}}\left(z_i^{\mathrm{lin}}(\bar{h}) > z_j^{\mathrm{lin}}(\bar{h})\right) < 1.$$

*Thus, unlike temperature scaling, the local affine surrogate of RSP randomizes token rankings at each decoding step.*

Proposition 1 is the minimal statement: pairwise order can flip. Exploration requires a stronger conclusion, namely that tokens outside the baseline candidate set can actually enter it.

**Proposition 2** (Top-$K$ entry under a nonempty surrogate entry region)**.** *Let $J_K(s_t)$ be the baseline top-$K$ set at decoding state $s_t$, and let $i \notin J_K(s_t)$. Define*

$$\mathcal{G}_{i,K} := \left\{\bar{h} \in \mathbb{R}^d : \#\left\{j \neq i : z_j^{\mathrm{lin}}(\bar{h}) > z_i^{\mathrm{lin}}(\bar{h})\right\} \leq K - 1\right\}.$$

*If $\mathcal{G}_{i,K}$ contains a nonempty open set and the centered prompt law assigns positive probability to every nonempty open subset of $\mathbb{R}^d$, then*

$$\Pr_{\bar{h}}\left(\bar{h} \in \mathcal{G}_{i,K}\right) > 0.$$

Proposition 2 is intentionally conditional: it does not claim that every excluded token admits such a top-$K$ entry region. It states that whenever the first-order model contains an open region in prompt space where a previously excluded token moves into the top-$K$, the centered perturbation reaches that region with positive probability. The open-region hypothesis can be stated in two equivalent ways. First, $\mathcal{G}_{i,K}$ is the union $\bigcup_S \mathcal{O}_S$ over subsets $S \subseteq \{j \neq i\}$ with $|S| \leq K-1$, where each $\mathcal{O}_S$ is the strict-inequality polyhedron in which precisely the indices in $S$ outrank token $i$ (Appendix E); each $\mathcal{O}_S$ is open in $\mathbb{R}^d$ as a finite intersection of open half-spaces, so $\mathcal{G}_{i,K}$ is nonempty whenever some $\mathcal{O}_S$ is nonempty. Second, sufficient strict-margin condition: the existence of any $\bar{h}_0 \in \mathbb{R}^d$ at which token $i$ outranks all but at most $K-1$ other tokens with a strict margin (no tie) yields $\bar{h}_0 \in \mathcal{O}_S$ for $S = \{j : z_j + a_j^\top \bar{h}_0 > z_i + a_i^\top \bar{h}_0\}$, and the open polyhedron $\mathcal{O}_S$ contains an open ball around $\bar{h}_0$. The complementary case—no such $\bar{h}_0$ exists—is exactly the situation in which token $i$ is dominated by the baseline top-$K$ in the affine logit surrogate, and Proposition 2 makes no claim there. Section 3.4 verifies empirically that nonempty entry regions exist for a substantial fraction of decoding states. Gaussianity is one sufficient way to realize the required support condition; temperature cannot reach $\mathcal{G}_{i,K}$ because it never changes token order at all. The mechanism here is branch opening, not mere probability flattening.

**Corollary 2** (Sample complexity of top-$K$ admission)**.** *Let $\mu_i := \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$ denote the per-sample admission probability for token $i$. Within $N$ independent fresh RSP draws, the probability of at least one admission is at least*

$$1 - (1-\mu_i)^N \;\geq\; 1 - \exp(-N\mu_i).$$

Corollary 2 closes the loop from per-sample positive probability to multi-trial exploration. Even when $\mu_i$ is small, $N \cdot \mu_i = \Theta(1)$ samples suffice to admit a previously excluded token with constant probability. The independence assumption first enters here: the single-run rank-flip and top-$K$ entry statements above do not require it. This is the formal bridge from the static guarantee of Proposition 2 to the trajectory-level Pass@$N$ phenomena reported in Section 3.4.

**From rank flips to trajectory diversity to accuracy gain.** Together, Propositions 1 and 2 close the logical chain from Section 3.3 to Section 3.4, up to the local-surrogate assumption above. Early RSP perturbations alter logit gaps before softmax, which can admit previously excluded tokens into the candidate set. Because autoregressive decoding compounds early choices, one such early admission is enough to push the trajectory into a different reasoning cluster even if later perturbations are already small. RSP therefore does not need to perturb the entire chain of thought uniformly: it only needs to alter the first few branching decisions before attention-mediated attenuation suppresses further noise, which is why early exploration and late stabilization coexist in the same mechanism.

The connection from trajectory diversity to accuracy gain requires a property of reasoning tasks themselves: the existence of multiple valid trajectories. We formalize this as a separate proposition rather than absorb it into the perturbation analysis.

**Proposition 3** (Exploration-meets-multi-path accuracy gain)**.** *Fix a problem $x$, and consider the protocol that runs the deterministic baseline once and additionally draws $N$ fresh RSPs $\bar{h}^{(1)}, \ldots, \bar{h}^{(N)}$, with the model decoded once per sample. Let $\mathcal{T}^\star(x)$ denote the (nonempty) set of trajectories reaching the correct answer and $p_{\mathrm{base}}(x) \in \{0,1\}$ the deterministic-baseline correctness indicator. Assume* (M1) *for each $n \in \{1, \ldots, N\}$, the marginal probability that the $n$-th RSP run produces a trajectory in $\mathcal{T}^\star(x)$ is at least $p_{\min}(x) > 0$ (the same lower bound for every $n$), and* (M2) *the $N$ RSP runs are mutually independent (independent prompt draws and, if decoding is stochastic, independent decoding randomness across runs). Then*

$$\Pr\big(\textit{Pass@}(N{+}1)\ \textit{correct on}\ x\big) \geq \max\big\{p_{\mathrm{base}}(x),\ 1-(1-p_{\min}(x))^N\big\} \geq 1 - e^{-Np_{\min}(x)}.$$

*The expected accuracy gain over the deterministic baseline is at least*

$$\frac{1}{|\mathcal{D}|}\sum_{x\in\mathcal{D}} \big(1-p_{\mathrm{base}}(x)\big)\big(1-\exp(-Np_{\min}(x))\big).$$

*Proof.* The Pass@$(N{+}1)$ event holds if the baseline run reaches $\mathcal{T}^\star(x)$ (probability $p_{\mathrm{base}}(x) \in \{0,1\}$) or if any of the $N$ RSP runs does. By (M2), the failure probabilities of the $N$ RSP runs multiply, so the probability that at least one RSP run reaches $\mathcal{T}^\star(x)$ is $1-(1-p_{\min}(x))^N$ by (M1). Taking the maximum gives the first lower bound; $1-x \leq e^{-x}$ gives the second. Restricting to $\{x : p_{\mathrm{base}}(x) = 0\}$ yields the expected gain. $\square$

Proposition 3 converts the local guarantees of Propositions 1 and 2 into a global accuracy claim, but only via the auxiliary assumption that $p_{\min}(x) > 0$. The next lemma formalizes a sufficient condition under which the perturbation theory of this section actually delivers $p_{\min}(x) > 0$, separating what is proved (positive measure of an entry region) from what is assumed (extension of a perturbed prefix to a correct completion).

**Lemma 2** (Bridge from open entry region to one-run success). *Specialize Proposition 2 to $K = 1$ (greedy decoding) and assume the same support condition on the centered prompt law. Suppose there exists a nonempty open ball $B \subseteq \mathbb{R}^d$ such that for every $\bar{h} \in B$ both of the following hold under a single RSP draw at the start of decoding:* (B1) *the RSP-induced trajectory under prompt $\bar{h}$ reaches some decoding state $s_{t^\star}(\bar{h})$ at which an excluded token $i^\star(\bar{h}) \notin J_1(s_{t^\star})$ satisfies $\bar{h} \in \mathcal{G}_{i^\star,1}(s_{t^\star})$, so greedy decoding selects $i^\star(\bar{h})$ at step $t^\star$;* (B2) *the resulting full trajectory belongs to $\mathcal{T}^\star(x)$. Then $p_{\min}(x) \geq \Pr(\bar{h} \in B) > 0$.*

*Proof.* By definition, $p_{\min}(x)$ is the per-sample probability that the RSP-induced full trajectory under one draw of $\bar{h}$ reaches $\mathcal{T}^\star(x)$. Conditions (B1) and (B2) state exactly that this event holds for every $\bar{h} \in B$. Since $B$ is a nonempty open subset of $\mathbb{R}^d$ and the centered prompt law assigns positive probability to every nonempty open set, $\Pr(\bar{h} \in B) > 0$, and the lower bound follows. □

Lemma 2 converts the abstract assumption $p_{\min}(x) > 0$ of Proposition 3 into the existence of an open ball $B$ satisfying two simultaneous properties: (B1) the perturbation theory of this section—specifically Proposition 2 at $K = 1$—supplies the open-set structure that triggers top-1 entry once a candidate decoding state $s_{t^\star}$ is fixed; (B2) the resulting full trajectory continues into a correct completion. The prefix-preservation aspect (the trajectory under $\bar{h}$ actually reaches a state $s_{t^\star}$ with the desired entry-region structure) and the completion property (B2) are both model- and dataset-dependent and treated as empirical hypotheses. Section 3.4 plays a load-bearing role here: it verifies that RSP-induced trajectories include both improvements and regressions over the baseline with net gain emerging under repeated sampling, which is precisely the empirical signature predicted by such an open ball $B$ existing for some but not all problems. The mechanism we have isolated is exploration meeting multi-path structure, not steering toward correctness.

Taken together, the causal chain is now explicit. Randomness matters because RSP is resampled, centered, and direction-agnostic; attention turns that perturbation into an early exploration coefficient and then anneals it; the resulting early rank flips open trajectories that temperature scaling alone cannot reach; and the multi-path structure of reasoning converts trajectory diversity into accuracy gain under repeated sampling. Section 3 therefore reports four observable faces of a single mechanism rather than four disconnected phenomena.

# 5 Conclusion

**Summary.** This work investigated the role of Random Soft Prompts (RSPs), continuous embeddings sampled from an isotropic Gaussian fitted to the empirical mean and variance of the pretrained embedding table and injected into the model input without any training. Through empirical and theoretical analyses, we established three findings. First, RSPs increase early-stage token entropy, top-1 probability reduction, and varentropy while the model's output stabilizes in later generation stages (Section 3). Second, RSPs induce stochastic token rankings and expand the effective support of the output distribution, a property that temperature sampling fundamentally lacks due to its ranking-preserving nature (Section 4). Third, combining RSPs with temperature sampling produces more diverse reasoning trajectories than either method alone, as measured by Pass@$n$ scaling (Section 3).

**Implications for GRPO training.** GRPO computes advantages by comparing multiple sampled trajectories within each group, making trajectory diversity essential for effective learning signals. Existing approaches such as DAPO [Yu et al., 2025] and CE-GPPO [Su et al., 2025] address entropy collapse by modifying the clipping range or reintroducing gradients from clipped tokens, but the underlying sampling process remains based on temperature sampling, which preserves token rankings and cannot expand the output support. Our theoretical and empirical results suggest that RSP can address this limitation at the sampling level: by perturbing the initial output distribution with freshly resampled random embeddings across compared rollouts, RSP provides a complementary source of diversity that is orthogonal to temperature-based exploration. We expect this to yield more effective learning signals for GRPO training, particularly in mitigating entropy collapse.

**Limitations.** Our experiments are conducted on mathematical reasoning benchmarks (MATH-500, GSM8K, AIME24) with a limited set of model families (Qwen2.5-Math, LLaMA-3.1). The energy-theoretic analysis in Section 4 relies on strong assumptions: the Hopfield–Attention equivalence (A1: $\mathbf{K} = \mathbf{V}$) and weight-tied layers (A2), which hold exactly only in Hopfield-type models and deep equilibrium models, respectively. Since real transformer architectures vary widely in their attention mechanisms and layer structures, the theoretical predictions are most directly applicable to models whose attention dynamics closely approximate the Hopfield energy framework. The GRPO application remains a theoretical expectation and has not been empirically validated.

**Future work.** Extending the experimental evaluation to other domains (e.g., code generation, commonsense reasoning) and larger model scales would test the generality of RSP's effects. Empirically validating the GRPO application by training with RSP-augmented rollouts and comparing against DAPO and CE-GPPO is a natural next step. Investigating the optimal RSP distribution and token count, which currently require per-model tuning, could lead to more principled injection strategies.

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## A Experimental Setup and Full Accuracy Breakdown

Table 6 provides the full per-position accuracy breakdown (Prefix, Infix, Suffix) corresponding to the best-across-positions results summarized in Table 1 in the main text. Table 7 lists the number of RSP tokens $L$ used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-Math-1.5B variants).

Table 6: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

| Model | Mode | MATH-500 | GSM8K | AIME24 |
|---|---|---|---|---|
| *Instruct Models* | | | | |
| LLaMA-3.1-8B-Instruct | Baseline | **52.20** | 85.75 | 6.67 |
| | Prefix | 45.00 (−7.2) | 76.35 (−9.4) | 3.33 (−3.3) |
| | Infix | 49.40 (−2.8) | 85.97 (+0.2) | **10.00** (+3.3) |
| | Suffix | 49.40 (−2.8) | **86.73** (+1.0) | **10.00** (+3.3) |
| Qwen2.5-Math-7B-Instruct | Baseline | 83.20 | 95.45 | 13.33 |
| | Prefix | 81.40 (−1.8) | 94.92 (−0.5) | 13.33 (+0.0) |
| | Infix | **84.20** (+1.0) | 95.60 (+0.2) | **16.67** (+3.3) |
| | Suffix | 83.40 (+0.2) | **95.68** (+0.2) | **16.67** (+3.3) |
| Qwen2.5-Math-1.5B-Instruct | Baseline | 73.00 | 85.29 | 10.00 |
| | Prefix | 74.80 (+1.8) | 85.29 (+0.0) | 13.33 (+3.3) |
| | Infix | **75.20** (+2.2) | **85.75** (+0.5) | 6.67 (−3.3) |
| | Suffix | 74.20 (+1.2) | 84.69 (−0.6) | **20.00** (+10.0) |
| *Base Models (Raw Text)* | | | | |
| Qwen2.5-Math-7B | Baseline | **72.20** | 84.15 | 6.67 |
| | Prefix | 71.60 (−0.6) | **84.31** (+0.2) | **16.67** (+10.0) |
| | Infix | 63.20 (−9.0) | 64.29 (−19.9) | **16.67** (+10.0) |
| | Suffix | 55.60 (−16.6) | 54.13 (−30.0) | 13.33 (+6.7) |
| Qwen2.5-Math-1.5B | Baseline | 61.40 | **77.86** | **16.67** |
| | Prefix | **64.40** (+3.0) | 74.30 (−3.6) | 10.00 (−6.7) |
| | Infix | 63.20 (+1.8) | 73.92 (−3.9) | 10.00 (−6.7) |
| | Suffix | 54.60 (−6.8) | 58.61 (−19.3) | 3.33 (−13.3) |
| *Base Models + ChatML (Format Mismatch)* | | | | |
| Qwen2.5-Math-7B | Baseline | 52.20 | 58.30 | **23.33** |
| | Prefix | 59.20 (+7.0) | 56.41 (−1.9) | 3.33 (−20.0) |
| | Infix | 62.00 (+9.8) | 69.07 (+10.8) | **23.33** (+0.0) |
| | Suffix | **70.40** (+18.2) | **75.97** (+17.7) | **23.33** (+0.0) |
| Qwen2.5-Math-1.5B | Baseline | 34.40 | 37.45 | 13.33 |
| | Prefix | **63.40** (+29.0) | **67.02** (+29.6) | **20.00** (+6.7) |
| | Infix | 39.20 (+4.8) | 50.42 (+13.0) | 6.67 (−6.7) |
| | Suffix | 51.00 (+16.6) | 50.64 (+13.2) | 13.33 (+0.0) |

Table 7: Number of RSP tokens ($L$) used for each model and benchmark.

| Model | MATH-500 | GSM8K | AIME24 |
|---|---|---|---|
| LLaMA-3.1-8B-Instruct | 15 | 20 | 15 |
| Qwen2.5-Math-7B-Instruct | 20 | 20 | 20 |
| Qwen2.5-Math-1.5B-Instruct | 20 | 10 | 20 |
| Qwen2.5-Math-7B | 10 | 10 | 20 |
| Qwen2.5-Math-1.5B | 10 | 10 | 20 |
| Qwen2.5-Math-1.5B (ChatML) | 20 | 20 | 10 |
| Qwen2.5-Math-7B (ChatML) | 20 | 20 | 20 |

### A.1 RSP Injection Positions and Prompt Templates

Below we show the prompt structure for each injection position across all model types. Blue text indicates RSP embeddings, and purple text indicates special tokens.

#### A.1.1 Prefix

RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

**ChatML (Qwen Instruct / Base + ChatML).**

```
[Random Embeddings (L tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

**LLaMA Chat Template.**

```
[Random Embeddings (L tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date:  December 2023
Today Date:  26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

**Raw Text (Qwen Base).**

```
[Random Embeddings (L tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{}.
```

#### A.1.2 Infix

A description text and special tokens are inserted into the prompt. The special token positions are then replaced with RSP embeddings at the embedding level.

**ChatML (Qwen Instruct / Base + ChatML).**

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning.  If these tokens are useful for your case, you can
use them as reference.  If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens:  <special_token_1>...<special_token_L>
<|im_end|>
<|im_start|>assistant
```

**LLaMA Chat Template.**

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date:  December 2023
Today Date:  26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
```

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens:
<reserved_special_token_0>...
<reserved_special_token_L>
<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

**Raw Text (Qwen Base).**

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens: <special_token_1>...<special_token_L>
Please reason step by step, and put your final answer within \boxed{}.
```

#### A.1.3 Suffix

For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token. For raw text models, RSP embeddings are concatenated at the end of the prompt.

**ChatML (Qwen Instruct / Base + ChatML).**

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}[Random Embeddings (L tokens)]<|im_end|>
<|im_start|>assistant
```

**LLaMA Chat Template.**

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}[Random Embeddings (L tokens)]<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

**Raw Text (Qwen Base).**

```
{question}
Please reason step by step, and put your final answer within \boxed{}.[Random
Embeddings (L tokens)]
```

### A.2 Answer Extraction and Grading

The final answer is extracted from the model output through a fallback chain. Each step is attempted in order; if extraction fails, the next step is tried.

**Answer Extraction Fallback Chain**

1. `"final answer is $...$.  I hope"` pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following `"the answer is"`
4. Text following `"final answer is"`
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing `$`, `\left`, `\right`, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose`, `rel_tol=1e-4`), and SymPy symbolic simplification.

### A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature $= 0$) and our unified answer extraction pipeline.

**TTSV.** Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

**LTPO.** Table 8 reports the per-benchmark hyperparameters.

Table 8: LTPO hyperparameters.

| Parameter | GSM8K | MATH-500 | AIME24 |
|---|---|---|---|
| tokens | 8 | 8 | 8 |
| steps | 20 | 20 | 20 |
| top_k | 10 | 10 | 10 |
| sigma | 20 | 20 | 5 |
| sigma_decay | 0.95 | 0.95 | 0.95 |
| lr | 1e-2 | 5e-3 | 5e-2 |

**SoftCoT.** Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

### A.4 Full Entropy Curves

Figure 5 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

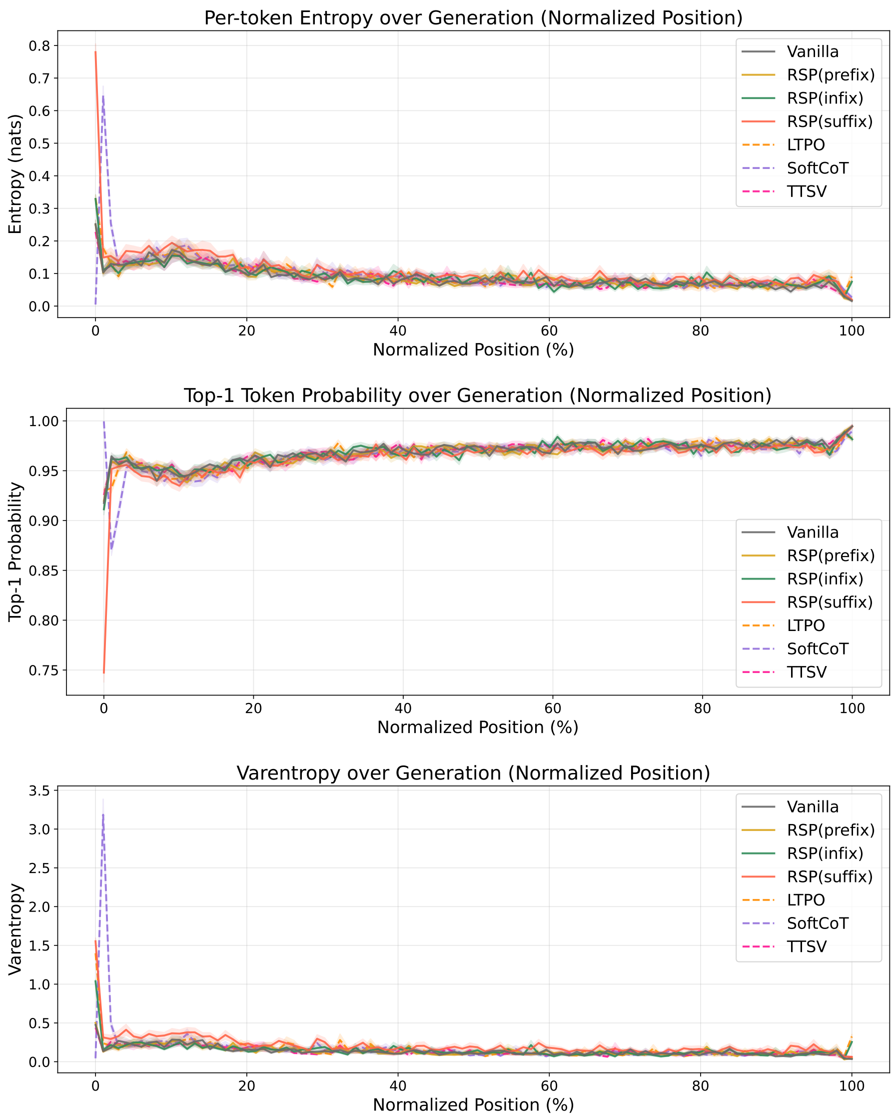



Figure 5: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates $\pm 1$ SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

# B Additional Semantic Diversity Analysis

This appendix complements the semantic diversity results reported in Section 3.4. We include per-problem t-SNE plots across a broader set of problems.

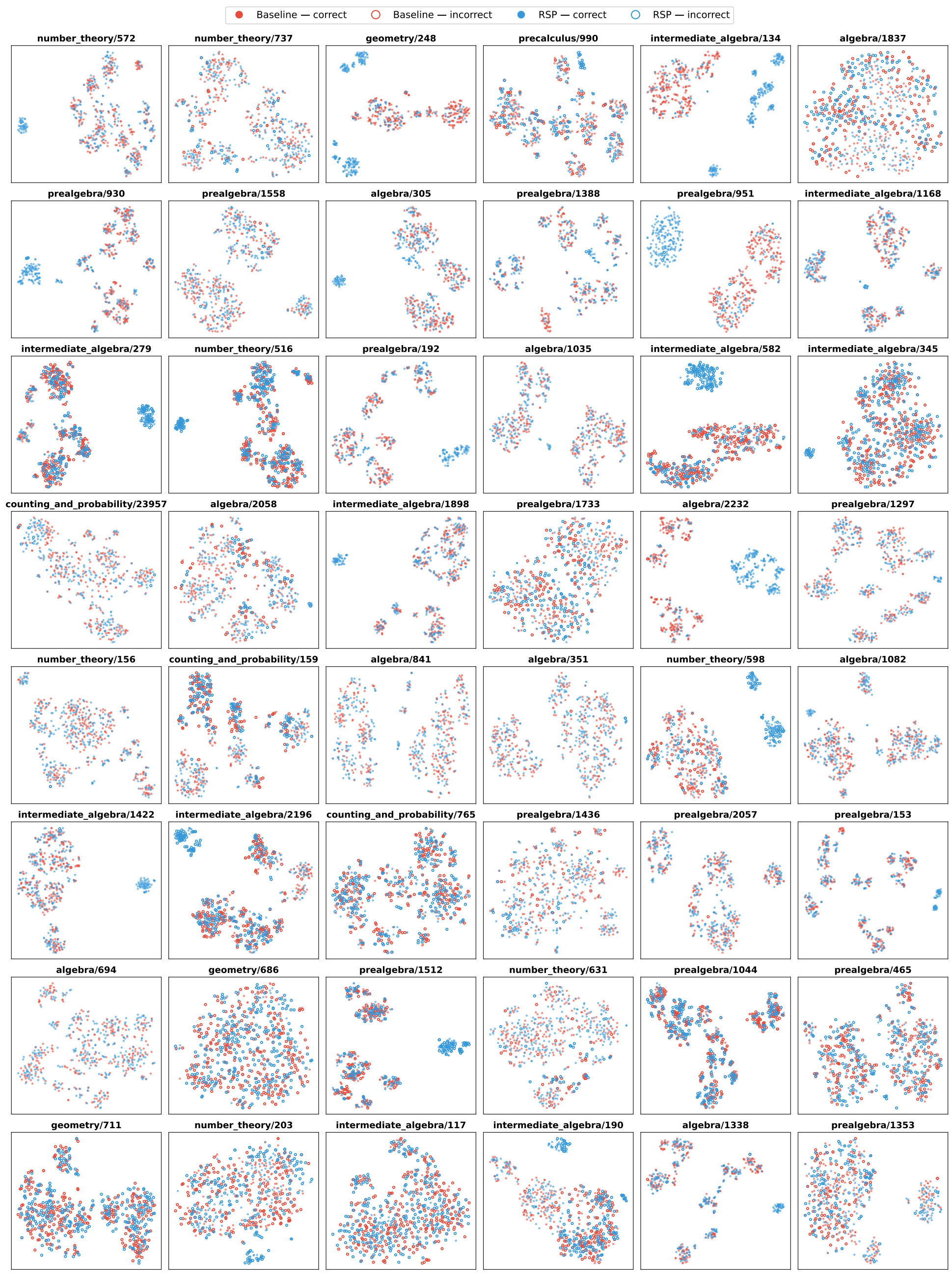



Figure 6: t-SNE projections of BGE-M3 embeddings for 48 out of the 64 evaluated MATH-500 problems (6×8 grid). Each subplot shows 300 baseline (red) and 300 RSP suffix (blue) samples under $\tau = 1.0$ with Qwen2.5-Math-1.5B-Instruct; filled markers denote correct answers and open markers denote incorrect. Subplot titles indicate `subject/problem_id`. Across most problems, RSP trajectories spread across multiple distinct regions while baseline samples concentrate in a single dominant cluster.

## C Token-Level Distribution Metrics

This appendix provides formal definitions for the three metrics used in the Token-level analysis of Section 3.4. Let $P_{\text{base}}$ denote the baseline next-token distribution at $\tau = 1.0$ and $P_{\text{target}}$ the candidate distribution being compared (e.g., the baseline at $\tau = 2.0$ or RSP at $\tau = 1.0$). All metrics are computed analytically from saved logits at the first generation step.

**Spearman rank correlation.** Spearman's $\rho$ is the Pearson correlation coefficient between the token ranks of two distributions:

$$\rho \;=\; \mathrm{Pearson}(\mathrm{rank}(P_{\text{target}}),\; \mathrm{rank}(P_{\text{base}}))\,. \tag{5}$$

Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore indicates rank reorganization beyond what temperature scaling can produce.

**Mass outside top-$K$.** Let $\mathrm{TopK}(P_{\text{base}})$ be the set of $K$ tokens with the highest probability under $P_{\text{base}}$. The probability mass that the candidate distribution places *outside* this set is

$$\mathrm{MassOutside}_K \;=\; 1 - \sum_{t \in \mathrm{TopK}(P_{\text{base}})} P_{\text{target}}(t). \tag{6}$$

This directly measures support expansion: how much probability the candidate assigns to tokens that are not in the baseline's preferred set. Reported values use $K = 10$.

**Jensen–Shannon divergence.** The symmetrized, bounded version of Kullback–Leibler divergence:

$$\mathrm{JS}(P, Q) \;=\; \tfrac{1}{2}\,\mathrm{KL}(P \,\|\, M) + \tfrac{1}{2}\,\mathrm{KL}(Q \,\|\, M), \qquad M = \tfrac{1}{2}(P + Q). \tag{7}$$

We use base-2 logarithms so that JS $\in [0, 1]$. Tokens absent from the stored top-100 are assigned a sentinel logit ($-10^9$) before softmax, which is negligible for our setting since the top-5 covers more than $99.9\%$ of the probability mass.

## D Why Random Prompts Are Semantically Neutral and Direction-Agnostic

This appendix formalizes the "why random" claim used in Section 4.1. The point is not that random prompts contain hidden task information. The point is structural: if the perturbation is meant to act as a generic exploration device rather than as a learned prompt prior, then it should be resampled across runs, centered so that it does not induce a fixed first-order drift, and direction-agnostic so that it does not under-serve whichever local branch direction matters for the current prompt.

### D.1 Semantic Neutrality

Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ and $\hat{h} := \bar{h}/\|\bar{h}\|$ whenever $\bar{h} \neq 0$. Let $u_1, \dots, u_M \in \mathbb{S}^{d-1}$ be any fixed unit vectors representing semantic anchor directions.

**Lemma 3** (Semantic neutrality)**.** *For any $\delta \in (0, 1)$,*

$$\Pr\left[\max_{r \in [M]} |\hat{h}^\top u_r| \leq \sqrt{\frac{2\log(2M/\delta)}{d-1}}\right] \geq 1 - \delta.$$

*Hence the normalized direction of an isotropic Gaussian prompt is simultaneously nearly orthogonal to every fixed finite collection of semantic anchors with high probability.*

*Proof.* Because $\bar{h}$ is isotropic Gaussian, its normalized direction $\hat{h}$ is uniformly distributed on the sphere $\mathbb{S}^{d-1}$. For a fixed unit vector $u \in \mathbb{S}^{d-1}$, the marginal $\hat{h}^\top u$ satisfies the standard spherical concentration bound [Vershynin, 2018, Theorem 3.4.6][Boucheron et al., 2013, Section 3.2],

$$\Pr\left(|\hat{h}^\top u| \geq t\right) \leq 2\exp\left(-\frac{(d-1)t^2}{2}\right) \qquad \text{for } t \in (0, 1),$$

which follows from the spherical-cap area bound $\sigma\{\hat{h} \in \mathbb{S}^{d-1} : \hat{h}^\top u \geq t\} \leq \frac{1}{2}(1 - t^2)^{(d-1)/2}$ together with $(1 - t^2)^{(d-1)/2} \leq \exp(-(d-1)t^2/2)$ derived from $\log(1 - x) \leq -x$ for $x \in [0, 1)$. Applying this bound to each $u_r$ and using a union bound,

$$\Pr\left[\max_{r \in [M]} |\hat{h}^\top u_r| \geq t\right] \leq 2M\exp\left(-\frac{(d-1)t^2}{2}\right).$$

Setting

$$t = \sqrt{\frac{2\log(2M/\delta)}{d-1}}$$

gives the claim. If the displayed value of $t$ exceeds 1, the statement is trivial because $|\hat{h}^\top u_r| \leq 1$ always. $\square$

The contrast with a token embedding is immediate. If $e_s \neq 0$ is a fixed token embedding and $\hat{e}_s := e_s/\|e_s\|$, then choosing the anchor direction $u = \hat{e}_s$ gives

$$|\hat{e}_s^\top u| = 1.$$

A token prompt is therefore maximally aligned with at least one semantic direction, whereas an isotropic random prompt is nearly orthogonal to every fixed finite set of such directions. This is the sense in which tokens inject semantic commitment while random prompts act as semantically uncommitted perturbations.

### D.2 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let $\Delta_{ij}$ be the baseline logit gap between tokens $i$ and $j$, and write the first-order surrogate as

$$\Delta_{ij}(\bar{h}) = \Delta_{ij} + b_{ij}^\top \bar{h}.$$

**Proposition 4** (No systematic first-order drift)**.** *If $\mathbb{E}[\bar{h}] = 0$, then*

$$\mathbb{E}\big[\Delta_{ij}(\bar{h})\big] = \Delta_{ij}$$

*for every token pair $(i, j)$. If a deterministic prompt $h_0$ is reused across runs, then*

$$\Delta_{ij}(h_0) - \Delta_{ij} = b_{ij}^\top h_0$$

*is a fixed directional bias.*

*Proof.* The centered case is immediate from linearity of expectation:

$$\mathbb{E}\big[\Delta_{ij}(\bar{h})\big] = \Delta_{ij} + b_{ij}^\top \mathbb{E}[\bar{h}] = \Delta_{ij}.$$

The deterministic case follows by substitution. □

The proposition does not say that a centered perturbation leaves every *realization* unchanged; it says there is no systematic first-order push toward one branch across runs. This is exactly the behavior desired from a generic exploration device. A fixed token or fixed soft prompt may still be useful, but it acts as a persistent prompt prior rather than as an unbiased exploration mechanism.

### D.3 Direction-Agnostic Coverage

The second requirement is directional coverage. Let $D$ be any zero-mean perturbation distribution in $\mathbb{R}^d$ with covariance $\Sigma_D$ and variance budget $\operatorname{tr}(\Sigma_D) \le \rho^2$. For a unit vector $b$, the first-order perturbation energy available along $b$ is $\operatorname{Var}_{h\sim D}(b^\top h) = b^\top \Sigma_D b$. This is the quantity that appears in the local linear logit surrogate used in the main text.

**Theorem 4** (Direction-agnostic coverage)**.** *Define*

$$\mathcal{E}(D) := \min_{\|b\|=1} \operatorname{Var}_{h\sim D}(b^\top h).$$

*Then*

$$\mathcal{E}(D) = \lambda_{\min}(\Sigma_D) \le \frac{\rho^2}{d},$$

*with equality if and only if* $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.

*Proof.* For any symmetric covariance matrix $\Sigma_D$,

$$\min_{\|b\|=1} b^\top \Sigma_D b = \lambda_{\min}(\Sigma_D),$$

so $\mathcal{E}(D) = \lambda_{\min}(\Sigma_D)$. Let the eigenvalues of $\Sigma_D$ be $\lambda_1, \ldots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \le \frac{1}{d}\sum_{m=1}^{d} \lambda_m = \frac{\operatorname{tr}(\Sigma_D)}{d} \le \frac{\rho^2}{d}.$$

Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. □

This theorem is a statement about second moments, not about Gaussianity by itself. Many distributions could realize isotropic covariance. We use a Gaussian in RSP because it is the simplest centered, rotation-symmetric family that matches the scale of pretrained embeddings. The theorem clarifies the structural advantage: a fixed prompt has $\Sigma_D = 0$ and therefore no across-run exploration, whereas any anisotropic perturbation leaves at least one branch-sensitive direction less explored than the isotropic optimum.

### D.4 Characterization Theorem

We formalize the claim that the centered isotropic Gaussian is the unique distribution satisfying all the structural desiderata for generic exploration.

**Theorem 5** (Characterization)**.** *Fix $\rho^2 > 0$ and $d \ge 1$. Among all probability distributions on $\mathbb{R}^d$ satisfying* (C1) *centered,* (C2) $\operatorname{tr}(\Sigma_D) \le \rho^2$*, and* (C3) $\mathcal{E}(D) = \rho^2/d$*, the unique distribution that maximizes differential entropy is $\mathcal{N}(0, (\rho^2/d)\mathbf{I}_d)$.*

*Proof.* Step 1 (covariance is forced). By Theorem 4, condition (C3) holds with equality if and only if $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. Therefore any distribution satisfying (C1)+(C2)+(C3) has mean zero and covariance exactly $(\rho^2/d)\mathbf{I}_d$.

Step 2 (Gaussianity is forced by maximum entropy). Among all distributions on $\mathbb{R}^d$ with prescribed mean 0 and prescribed covariance $\Sigma$, the maximum-entropy distribution is the Gaussian $\mathcal{N}(0, \Sigma)$ [Cover and Thomas, 2006, Theorem 12.1.1], and the maximizer is unique. Applying this with $\Sigma = (\rho^2/d)\mathbf{I}_d$ gives the result. □

**Why this characterization is meaningful.** Step 1 says any distribution satisfying the second-moment desiderata must have isotropic covariance scaled to the budget; this immediately rules out PCA-aligned noise, vocabulary token sampling, and any reused soft prompt at the second-moment level. Step 2 says among the remaining (continuous, isotropic) distributions, choosing Gaussian is the unique distributional choice that adds no further structural assumption beyond the prescribed second moments. RSP can therefore be described as the minimum-information distribution consistent with the second-moment desiderata.

**Proposition 5** (Covariance pushforward under the value projection)**.** *Let $\bar{h}$ be a centered random vector with covariance $\sigma_E^2 \mathbf{I}_d$, and let $v = W\bar{h}$ for a matrix $W \in \mathbb{R}^{d_v \times d}$. Then*

$$\mathrm{Cov}(v) = \sigma_E^2 W W^\top.$$

*Consequently, for every unit vector $u \in \mathbb{S}^{d_v - 1}$,*

$$\sigma_E^2 \lambda_{\min}(WW^\top) \le \mathrm{Var}(u^\top v) = \sigma_E^2 u^\top W W^\top u \le \sigma_E^2 \lambda_{\max}(WW^\top).$$

*In particular, if $W$ is full rank then the projected perturbation has strictly positive variance in every output-space direction.*

*Proof.* Since $v = W\bar{h}$ and $\mathbb{E}[\bar{h}] = 0$,

$$\mathrm{Cov}(v) = \mathbb{E}[vv^\top] = W\,\mathbb{E}[\bar{h}\bar{h}^\top]\,W^\top = \sigma_E^2 W W^\top.$$

For any unit vector $u$,

$$\mathrm{Var}(u^\top v) = u^\top \mathrm{Cov}(v)\,u = \sigma_E^2 u^\top W W^\top u,$$

and the Rayleigh quotient bounds give the stated inequalities. $\square$

## E Proofs for Attention-Mediated Annealing

### E.1 Hopfield-Energy Descent and Basin Trapping

We restate the Modern Hopfield correspondence in the form used in the main text. For a key matrix $\mathbf{K} = (\mathbf{k}_1, \ldots, \mathbf{k}_n)$ at a fixed query position, the Hopfield energy is

$$E_{\mathbf{K}}(\sigma) := -\beta^{-1} \log \sum_{j=1}^{n} \exp(\beta\, \mathbf{k}_j^{\top} \sigma) + \tfrac{1}{2}\|\sigma\|^2.$$

A direct calculation gives the gradient

$$\nabla E_{\mathbf{K}}(\sigma) = \sigma - \mathbf{K}^{\top}\mathrm{softmax}(\beta \mathbf{K}\sigma).$$

Under (A1) $\mathbf{K} = \mathbf{V}$, the attention update with this $\mathbf{K}$ is therefore $\sigma^{\text{new}} = \sigma - \nabla E_{\mathbf{K}}(\sigma) = \mathbf{K}^{\top}\mathrm{softmax}(\beta \mathbf{K}\sigma)$, which is one gradient step on $E_{\mathbf{K}}$ at unit step size [Ramsauer et al., 2021].

**Lemma 4** (Descent inequality)**.** *If $E_{\mathbf{K}}$ is $L_E$-smooth on a neighborhood of $\sigma$, then the unit-step gradient update $\sigma^{new} := \sigma - \nabla E_{\mathbf{K}}(\sigma)$ satisfies*

$$E_{\mathbf{K}}(\sigma^{new}) \le E_{\mathbf{K}}(\sigma) - \big(1 - \tfrac{L_E}{2}\big)\|\nabla E_{\mathbf{K}}(\sigma)\|^2.$$

*Proof.* The standard $L_E$-smooth quadratic upper bound [Nesterov, 2018, Lemma 1.2.3] states $E(\sigma + v) \le E(\sigma) + \nabla E(\sigma)^{\top} v + (L_E/2)\|v\|^2$ for any $v$ in the smoothness neighborhood. Applying this to $v = -\nabla E(\sigma)$ gives $E(\sigma - \nabla E) \le E(\sigma) - \|\nabla E\|^2 + (L_E/2)\|\nabla E\|^2 = E(\sigma) - (1 - L_E/2)\|\nabla E\|^2$. □

**Theorem 6** (Deterministic descent and sublevel-set invariance)**.** *Assume* (A1) $\mathbf{K} = \mathbf{V}$, (A2) *the attention component is weight-tied across blocks, and* (A3) $E_{\mathbf{K}}$ *is $L_E$-smooth on a neighborhood of the unperturbed trajectory with $L_E < 2$. Then the unit-step iterate $\sigma^{(l+1)} = \sigma^{(l)} - \nabla E_{\mathbf{K}}(\sigma^{(l)})$ is strictly monotone-descending in $E_{\mathbf{K}}$ at every non-stationary point, and the sublevel set $\{\sigma : E_{\mathbf{K}}(\sigma) \le E_{\mathbf{K}}(\sigma^{(0)})\}$ is forward-invariant.*

*Proof.* By Lemma 4, $E_{\mathbf{K}}(\sigma^{(l+1)}) \le E_{\mathbf{K}}(\sigma^{(l)}) - (1 - L_E/2)\|\nabla E_{\mathbf{K}}(\sigma^{(l)})\|^2$. Under (A3), the coefficient $1 - L_E/2 > 0$, so $E_{\mathbf{K}}(\sigma^{(l+1)}) \le E_{\mathbf{K}}(\sigma^{(l)})$ with strict inequality unless $\nabla E_{\mathbf{K}}(\sigma^{(l)}) = 0$, which proves forward invariance. □

**Corollary 3** (Connected-component containment for the unit-step iterate)**.** *Let $S := \{\sigma : E_{\mathbf{K}}(\sigma) \le E_{\mathbf{K}}(\sigma^{(0)})\}$ and let $C(\sigma^{(0)})$ denote the connected component of $S$ containing $\sigma^{(0)}$. Define*

$$\mathrm{sep}(C(\sigma^{(0)})) \;:=\; \inf\big\{\|x - y\| \,:\, x \in C(\sigma^{(0)}),\ y \in S \setminus C(\sigma^{(0)})\big\},$$

*with the convention* $\mathrm{sep} = +\infty$ *if $S$ is connected. Assume* $\mathrm{sep}(C(\sigma^{(0)})) > 0$. *Then the unit-step iterate $\sigma^{(l+1)} = \sigma^{(l)} - \nabla E_{\mathbf{K}}(\sigma^{(l)})$ remains in $C(\sigma^{(0)})$ for all $l \ge 0$ whenever*

$$\|\nabla E_{\mathbf{K}}(\sigma^{(l)})\| \;<\; \mathrm{sep}(C(\sigma^{(0)})) \qquad \textit{for every } l.$$

*Proof.* By Theorem 6, every iterate lies in $S$. Suppose by induction $\sigma^{(l)} \in C(\sigma^{(0)})$. The displacement $\sigma^{(l+1)} - \sigma^{(l)} = -\nabla E_{\mathbf{K}}(\sigma^{(l)})$ has Euclidean length $\|\nabla E_{\mathbf{K}}(\sigma^{(l)})\|$. A jump to a distinct component $C' \subseteq S$ would require this length to be at least $\inf\{\|x - y\| : x \in C(\sigma^{(0)}),\, y \in C'\} \ge \mathrm{sep}(C(\sigma^{(0)}))$, which the hypothesis excludes. Hence $\sigma^{(l+1)} \in C(\sigma^{(0)})$. □

**Positivity of** $\mathrm{sep}$**.** $\mathrm{sep}(C) > 0$ holds whenever the components of $S$ are spatially separated by a region of strictly higher energy than $E_{\mathbf{K}}(\sigma^{(0)})$, which we treat as a problem-dependent assumption rather than a structural property: it requires that the relevant components of the Hopfield energy do not share closure boundary points at level $E_{\mathbf{K}}(\sigma^{(0)})$. If $\mathrm{sep} = 0$ the corollary becomes vacuous; in that case the trajectory is still in $S$ by Theorem 6, but is not formally trapped in $C(\sigma^{(0)})$.

**Caveats and scope.** Theorem 6 depends on three assumptions that are individually mild but jointly idealized for real transformers. (A1) $\mathbf{K} = \mathbf{V}$ collapses the key and value projections; this is the standard Modern Hopfield assumption [Ramsauer et al., 2021] and isolates the energy-iteration aspect of attention from the additional value-projection degree of freedom. (A2) weight-tying treats the attention component as a single repeated map; without it, each block has its own energy and the trajectory does not minimize a single fixed function across blocks. (A3) is a smoothness condition on $E_{\mathbf{K}}$. For the Hopfield energy used here, $\nabla^2 E_{\mathbf{K}}(\sigma) = \mathbf{I} - \beta\mathbf{K}^\top(\mathrm{diag}(p_\sigma) - p_\sigma p_\sigma^\top)\mathbf{K}$ where $p_\sigma = \mathrm{softmax}(\beta\mathbf{K}\sigma)$, so the operator-norm bound on $\nabla^2 E_{\mathbf{K}}$ depends on the inverse-temperature $\beta$ and on $\|\mathbf{K}\|$. (A3) holds whenever $\beta\|\mathbf{K}\|^2$ is sufficiently small, which is the standard regime in which the Hopfield iterate converges in a small number of steps. The theorem therefore characterizes an idealized tendency of attention-only iteration in this regime; it is not a literal property of every transformer block. In particular, sublevel-set confinement follows from Theorem 6, whereas connected-component confinement additionally requires the separation and per-step gradient assumptions of Corollary 3. Its role in the main text is to supply an idealized source of exploration deficit that an injected stochastic perturbation addresses; the empirical link is the cross-layer attention decay observed in Figure 4 and the trajectory diversity gain reported in Section 3.4.

### E.2 Exact Attention Decomposition and Magnitude Control

**Proposition 6** (Attention decomposition)**.** *Let $\alpha_{ij}^{(l)}$ be the attention weights for query position $i$ at layer $l$, with $n$ real tokens and $k$ random tokens. Define*

$$w_{r,i}^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)}$$

*and, for $j \in [n]$,*

$$\tilde{\alpha}_{ij}^{(l)} := \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}}.$$

*Then*

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)})\,\tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)},$$

*where*

$$\tilde{\mathbf{x}}_i^{(l+1)} := \sum_{j=1}^{n} \tilde{\alpha}_{ij}^{(l)} \mathbf{v}_j^{(l)} \qquad \text{and} \qquad \boldsymbol{\eta}_i^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}.$$

*Proof.* The attention output is

$$\mathbf{x}_i^{(l+1)} = \sum_{j=1}^{n} \alpha_{ij}^{(l)} \mathbf{v}_j^{(l)} + \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}.$$

By definition, $\sum_{j=1}^{n} \alpha_{ij}^{(l)} = 1 - w_{r,i}^{(l)}$, so

$$\sum_{j=1}^{n} \alpha_{ij}^{(l)} \mathbf{v}_j^{(l)} = (1 - w_{r,i}^{(l)}) \sum_{j=1}^{n} \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}} \mathbf{v}_j^{(l)} = (1 - w_{r,i}^{(l)})\,\tilde{\mathbf{x}}_i^{(l+1)}.$$

Substituting this identity gives the decomposition. □

**Corollary 4** (Magnitude scales with random-token attention)**.** *For every realization of the random values,*

$$\|\boldsymbol{\eta}_i^{(l)}\| \leq w_{r,i}^{(l)} \max_{j\in[k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

*Proof.* By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(l)}\| = \left\| \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)} \right\| \leq \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \|\mathbf{v}_{r_j}^{(l)}\| \leq w_{r,i}^{(l)} \max_{j\in[k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

□

Corollary 4 is the only quantitative fact needed in the main text. No independence assumption between attention weights and random keys is required. Once $w_{r,i}^{(l)}$ is small, the random contribution must be small as well.

### E.3 Attention Decay Bound

**Theorem 7** (Attention decay)**.** *The total attention on random tokens satisfies*

$$w_{r,i}^{(l)} \leq \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}, \tag{8}$$

*where* $\Delta_i^{(l)} := \min_{j\in[n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j'\in[k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ *is the real-vs-random logit gap.*

*Proof.* Let $s_j = \mathbf{q}_i^\top \mathbf{k}_j/\sqrt{d}$ for real keys and $s_{r_{j'}} = \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}/\sqrt{d}$ for random keys. By definition of $\Delta_i^{(l)}$, each real key satisfies $s_j \geq \Delta_i^{(l)}/\sqrt{d} + s_{r,\max}$ where $s_{r,\max} = \max_{j'} s_{r_{j'}}$. Therefore $\exp(s_j) \geq \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max})$ for every real key. Summing over $n$ real keys gives

$$\sum_{j=1}^{n} \exp(s_j) \geq n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max}) \geq n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \frac{1}{k} \sum_{j'=1}^{k} \exp(s_{r_{j'}}).$$

Let $R := \sum_{j'=1}^{k} \exp(s_{r_{j'}})$. Then

$$w_{r,i}^{(l)} = \frac{R}{\sum_{j=1}^{n} \exp(s_j) + R} \leq \frac{R}{n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot R/k + R} = \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}.$$

□

**Corollary 5** (Sequence-wise annealing under a fixed logit gap)**.** *For fixed $k$ and $\Delta_i^{(l)}$, the upper bound*

$$f(n) = \frac{k}{k + n \exp(\Delta_i^{(l)}/\sqrt{d})}$$

*is strictly decreasing in $n$. Consequently, cache growth alone decreases the maximum perturbation mass compatible with that gap.*

*Proof.* Differentiate $f(n)$ with respect to $n$:

$$f'(n) = -\frac{k \exp(\Delta_i^{(l)}/\sqrt{d})}{\left(k + n \exp(\Delta_i^{(l)}/\sqrt{d})\right)^2} < 0.$$

Thus $f(n)$ is strictly decreasing. □

**Corollary 6** (Vanishing random contribution under annealing)**.** *If $n \to \infty$ along decoding while $k$ and $\Delta_i^{(l)}$ remain fixed, then $w_{r,i}^{(l)} \to 0$. Moreover, for any fixed realization of the $k$ random values,*

$$\|\boldsymbol{\eta}_i^{(l)}\| \leq w_{r,i}^{(l)} \max_{j\in[k]} \|\mathbf{v}_{r_j}^{(l)}\| \to 0.$$

*Since the implementation samples finite-dimensional vectors and $k$ is finite, the displayed maximum is almost surely finite.*

*Proof.* Corollary 5 implies $w_{r,i}^{(l)} \to 0$. The bound on $\|\boldsymbol{\eta}_i^{(l)}\|$ is Corollary 4. Since the multiplier $\max_{j\in[k]} \|\mathbf{v}_{r_j}^{(l)}\|$ is finite for any fixed realization of finitely many sampled values, the product converges to zero. □

### E.4 Feed-Forward Propagation under Local Lipschitz Control

A transformer block applies an attention sublayer followed by a feed-forward sublayer with residual connections:

$$\mathbf{x}_i^{(l,\text{attn})} = \mathbf{x}_i^{(l)} + \text{Attn}^{(l)}(\mathbf{x}_i^{(l)}), \qquad \mathbf{x}_i^{(l+1)} = \mathbf{x}_i^{(l,\text{attn})} + F^{(l)}(\mathbf{x}_i^{(l,\text{attn})}).$$

Let $\mathbf{x}_i^{(l,\text{attn,RSP})}$ and $\mathbf{x}_i^{(l,\text{attn,base})}$ denote the post-attention states under RSP and the unperturbed forward pass at the same input $\mathbf{x}_i^{(l)}$.

**Lemma 5** (Per-block FFN propagation)**.** *The post-attention difference satisfies*

$$\|\mathbf{x}_i^{(l,attn,RSP)} - \mathbf{x}_i^{(l,attn,base)}\| \leq \|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|.$$

*If, in addition, $F^{(l)}$ is $L^{(l)}$-Lipschitz on a bounded set containing both $\mathbf{x}_i^{(l,attn,RSP)}$ and $\mathbf{x}_i^{(l,attn,base)}$, then*

$$\|\mathbf{x}_i^{(l+1,RSP)} - \mathbf{x}_i^{(l+1,base)}\| \leq (1+L^{(l)})\big(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|\big).$$

*Proof.* The unperturbed attention sublayer produces $\mathbf{x}_i^{(l,\text{attn,base})} = \mathbf{x}_i^{(l)} + \tilde{\mathbf{x}}_i^{(l+1)}$, while the RSP attention sublayer produces $\mathbf{x}_i^{(l,\text{attn,RSP})} = \mathbf{x}_i^{(l)} + (1-w_{r,i}^{(l)})\tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)}$ by Proposition 6. Subtracting,

$$\mathbf{x}_i^{(l,\text{attn,RSP})} - \mathbf{x}_i^{(l,\text{attn,base})} = \boldsymbol{\eta}_i^{(l)} - w_{r,i}^{(l)}\tilde{\mathbf{x}}_i^{(l+1)},$$

and the triangle inequality gives the first bound. For the second bound, write the post-FFN difference as

$$\big[\mathbf{x}_i^{(l,\text{attn,RSP})} + F^{(l)}(\mathbf{x}_i^{(l,\text{attn,RSP})})\big] - \big[\mathbf{x}_i^{(l,\text{attn,base})} + F^{(l)}(\mathbf{x}_i^{(l,\text{attn,base})})\big].$$

By Lipschitz continuity, $\|F^{(l)}(\mathbf{x}^{(l,\text{attn,RSP})}) - F^{(l)}(\mathbf{x}^{(l,\text{attn,base})})\| \leq L^{(l)}\|\mathbf{x}_i^{(l,\text{attn,RSP})} - \mathbf{x}_i^{(l,\text{attn,base})}\|$. The triangle inequality combined with the first bound gives the second. $\square$

**Corollary 7** (Cross-block propagation control)**.** *Define $\Delta^{(l)} := \mathbf{x}_i^{(l,RSP)} - \mathbf{x}_i^{(l,base)}$ with $\Delta^{(0)} = \mathbf{0}$ (matched inputs at layer 0), and let $L_{block}^{(l)}$ denote the local Lipschitz constant of the block map $B_{base}^{(l)} : \mathbf{x} \mapsto \mathbf{x} + \text{Attn}^{(l)}(\mathbf{x}) + F^{(l)}(\mathbf{x} + \text{Attn}^{(l)}(\mathbf{x}))$ at layer $l$. Then*

$$\|\Delta^{(l+1)}\| \leq (1 + L_{block}^{(l)})\,\|\Delta^{(l)}\| + (1+L^{(l)})\big(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|\big).$$

*Iterating across $L$ blocks with $\Delta^{(0)} = \mathbf{0}$ yields*

$$\|\Delta^{(L+1)}\| \leq \sum_{l=0}^{L}\prod_{l'=l+1}^{L}(1+L_{block}^{(l')})\cdot(1+L^{(l)})\big(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|\big).$$

*Proof.* Decompose the cross-block discrepancy at layer $l+1$ into two contributions:

$$\Delta^{(l+1)} = \underbrace{B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l,\text{RSP})}) - B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l,\text{base})})}_{\text{(I) propagation of prior } \Delta^{(l)}} + \underbrace{B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l,\text{base})}) - B_{\text{base}}^{(l)}(\mathbf{x}^{(l,\text{base})})}_{\text{(II) RSP-induced perturbation injected at block } l},$$

where $B_{\text{RSP}}^{(l)}$ replaces $\text{Attn}^{(l)}$ with the RSP-augmented attention. Term (I) is bounded via the block Lipschitz constant: $\|(I)\| \leq L_{\text{block}}^{(l)}\|\Delta^{(l)}\|$. With residuals included one gets the factor $(1+L_{\text{block}}^{(l)})$ when also propagating the residual term $\Delta^{(l)}$ itself. Term (II) is exactly the per-block injection from Lemma 5, bounded by $(1+L^{(l)})\big(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|\big)$. Combining via the triangle inequality gives the recursion. The summed bound follows by unrolling the recursion: with $\Delta^{(0)} = \mathbf{0}$, induction on $L$ shows that $\|\Delta^{(L+1)}\|$ is at most the displayed sum. $\square$

**Boundedness of $L^{(l)}$ for standard architectures.** Modern transformer blocks combine LayerNorm, GELU or SwiGLU activations, and linear projections. LayerNorm has Lipschitz constant on the order of $\|\gamma\|/\inf\|x\|$ on every set bounded away from the origin, where $\gamma$ is the LayerNorm scale parameter [Xiong et al., 2020]. GELU and SwiGLU are smooth and therefore locally Lipschitz on bounded activation regions, and linear projections have Lipschitz constant equal to the spectral norm of the weight matrix. The composed self-attention sublayer has a Lipschitz constant that depends on the spectral norms of $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ and the inverse-temperature $\beta$ as analyzed by Kim et al. [2021]. On the bounded activation regions traversed during a single forward pass, the per-block Lipschitz constants in Corollary 7 are therefore finite. For a fixed model depth this yields a finite cumulative bound on $\|\Delta^{(L+1)}\|$; stronger depth-uniform non-expansion would require additional contraction assumptions not imposed here. This is the formal sense in which the injected perturbation stays controlled across a finite forward pass, while the size of each new injection is set by $w_{r,i}^{(l)}$.

### E.5 Per-Block Basin Escape Probability

We formalize the directional escape bound stated in the main text.

We work at the level of a single block, where the random contribution $\boldsymbol{\eta}_i^{(l)}$ is exactly Gaussian by construction.

**Lemma 6** (Directional variance of the per-block injection)**.** *Conditioned on the attention weights $\alpha^{(l)}$, $\boldsymbol{\eta}_i^{(l)}$ is Gaussian with covariance $\sigma_E^2\|\alpha_R^{(l)}\|^2\mathbf{I}_d$, where $\alpha_R^{(l)} := (\alpha_{i,n+1}^{(l)}, \ldots, \alpha_{i,n+k}^{(l)})$ collects the random-token weights and $\sigma_E^2$ is the per-coordinate variance of the embedding distribution. In particular, for every unit direction $u$,*

$$\mathrm{Var}(u^\top \boldsymbol{\eta}_i^{(l)} \mid \alpha^{(l)}) = \sigma_E^2\|\alpha_R^{(l)}\|^2 \;\geq\; \frac{\sigma_E^2 (w_{r,i}^{(l)})^2}{k},$$

*the lower bound following from $\sum_j (\alpha_{i,n+j}^{(l)})^2 \geq (\sum_j \alpha_{i,n+j}^{(l)})^2/k = (w_{r,i}^{(l)})^2/k$ by Cauchy–Schwarz.*

*Proof.* The random values are independent and isotropic Gaussian with covariance $\sigma_E^2\mathbf{I}_d$. Linear combinations of independent Gaussians are Gaussian, with covariance equal to the sum of squared coefficients times the per-component variance. The Cauchy–Schwarz lower bound on $\|\alpha\|^2$ uses non-negativity of the weights and $\sum_j \alpha_j = w_{r,i}^{(l)}$. $\square$

**Proposition 7** (Per-block directional escape probability)**.** *Let $\sigma^{(l,\text{post-attn,base})}$ denote the unperturbed post-attention state at block $l$, and suppose it lies in a basin $B$ whose boundary along a unit direction $u$ is at distance $r_u^{(l)} > 0$. The post-attention RSP state differs from the unperturbed one by $\boldsymbol{\eta}_i^{(l)} - w_{r,i}^{(l)}\tilde{\mathbf{x}}_i^{(l+1)}$ (Lemma 5). Conditioning on $\alpha^{(l)}$, the random component $u^\top\boldsymbol{\eta}_i^{(l)}$ is Gaussian with variance $\sigma_E^2\|\alpha_R^{(l)}\|^2 > 0$, hence*

$$\Pr\Big(u^\top\boldsymbol{\eta}_i^{(l)} > r_u^{(l)} + w_{r,i}^{(l)}\, u^\top\tilde{\mathbf{x}}_i^{(l+1)} \,\Big|\, \alpha^{(l)}\Big) \;\geq\; \Phi\left(-\frac{r_u^{(l)} + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|}{\sigma_E\|\alpha_R^{(l)}\|}\right) \;>\; 0.$$

*This event is sufficient for the perturbed post-attention state to cross the boundary of $B$ along $u$.*

*Proof.* By Lemma 6, $u^\top\boldsymbol{\eta}_i^{(l)} \sim \mathcal{N}(0, \sigma_E^2\|\alpha_R^{(l)}\|^2)$ conditional on $\alpha^{(l)}$ (the centering follows from $\mu_E$ being absorbed by the centered-prompt construction in Section 4.1). The post-attention difference along $u$ is $u^\top\boldsymbol{\eta}_i^{(l)} - w_{r,i}^{(l)}u^\top\tilde{\mathbf{x}}_i^{(l+1)}$, so crossing the boundary along $u$ requires this quantity to exceed $r_u^{(l)}$. Solving and applying the Gaussian one-sided tail $\Pr(\mathcal{N}(0,s^2) > t) = 1 - \Phi(t/s) = \Phi(-t/s)$ gives the bound. The bound is positive whenever the variance $\sigma_E^2\|\alpha_R^{(l)}\|^2$ is positive, which holds whenever any random-token weight is nonzero. $\square$

**Corollary 8** (Direction-agnostic positive escape)**.** *Suppose $w_{r,i}^{(l)} > 0$. Then for every unit direction $u$ with $r_u^{(l)} > 0$, the probability of crossing the basin boundary along $u$ within block $l$ is bounded below* *by*

$$\Phi\left(-\frac{(r_u^{(l)} + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|)\sqrt{k}}{\sigma_E\, w_{r,i}^{(l)}}\right) > 0,$$

*where the dependence on $u$ enters through $r_u^{(l)}$.*

*Proof.* Apply Proposition 7 with the lower bound $\|\alpha_R^{(l)}\|^2 \geq (w_{r,i}^{(l)})^2/k$ from Lemma 6. $\square$

**Caveats and scope.** The per-block directional escape bound certifies that RSP perturbation has positive probability of crossing every barrier-orthogonal direction with positive boundary distance, simultaneously. It depends on three localizations. First, the bound is for the post-attention state at a single block; whether the cumulative state at the end of the block (after FFN and residuals) also crosses the boundary follows from Lipschitz continuity (Lemma 5), but the basin geometry at the post-block state need not coincide with that at the post-attention state. Second, the bound treats the basin $B$ as a property of the energy at this block; for cross-block compounding, one must additionally assume a degree of basin compatibility across blocks, which we do not formalize. Third, the qualitative claim that RSP-induced trajectories settle in different reasoning clusters than the baseline (Section 3.4) is empirical, not proven as a quantitative escape rate.

### E.6 Proofs for Output Distribution Shift

The main text makes two output-level claims. Proposition 1 establishes pairwise rank randomness, and Proposition 2 upgrades that statement to top-$K$ entry. Both are statements about a local first-order surrogate of the perturbed logits.

#### E.6.1 Autoregressive Formulation

In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^{T} p(y_t \mid x, y_{<t}),$$

where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single global distribution.

#### E.6.2 Local Linearization of Logits under RSP

Let $\bar{h}$ denote the centered random soft prompt. We model the perturbed logits as a function $z_i(\bar{h})$, assuming local smoothness in a neighborhood of $\bar{h} = 0$:

$$z_i(\bar{h}) = z_i(0) + \nabla_h z_i(0)^\top \bar{h} + r_i(\bar{h}),$$

where $r_i(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $a_i := \nabla_h z_i(0)$, we obtain the local affine surrogate

$$z_i^{\mathrm{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}.$$

This approximation is valid for sufficiently small $\|\bar{h}\|$ and is used as a first-order surrogate for the direction of perturbation rather than as an exact global model of the transformer logits.

For the two propositions below we isolate the only support property that is actually used. Let $D$ denote the centered prompt law. We assume

$$\text{(S1)} \qquad D(U) > 0 \quad \text{for every nonempty open set } U \subseteq \mathbb{R}^d.$$

The Gaussian instantiation in Section 4.1 satisfies (S1), but the proofs below require only this support condition.

### E.6.3 Proof of Proposition 1 (Stochastic Ranking)

*If there exists a token pair $(i, j)$ such that $a_i \neq a_j$, and if the centered prompt law satisfies (S1), then* $0 < \Pr_{\bar{h}}(z_i^{\text{lin}}(\bar{h}) > z_j^{\text{lin}}(\bar{h})) < 1$.

*Proof.* Under the local affine surrogate,

$$z_i^{\text{lin}}(\bar{h}) - z_j^{\text{lin}}(\bar{h}) = (z_i - z_j) + (a_i - a_j)^\top \bar{h}.$$

Let $b_{ij} := a_i - a_j \neq 0$. The perturbed logit difference

$$\Delta_{ij}(\bar{h}) := (z_i - z_j) + b_{ij}^\top \bar{h}$$

vanishes on the affine hyperplane

$$H_0 := \{\bar{h} \in \mathbb{R}^d : (z_i - z_j) + b_{ij}^\top \bar{h} = 0\}.$$

Because $b_{ij} \neq 0$, $H_0$ has codimension one and partitions $\mathbb{R}^d$ into the two nonempty open half-spaces

$$H_+ := \{\bar{h} : (z_i - z_j) + b_{ij}^\top \bar{h} > 0\}, \qquad H_- := \{\bar{h} : (z_i - z_j) + b_{ij}^\top \bar{h} < 0\}.$$

By (S1), both $H_+$ and $H_-$ have positive probability, which yields

$$0 < \Pr_{\bar{h}}(\Delta_{ij}(\bar{h}) > 0) < 1.$$

□

### E.6.4 Proof of Proposition 2 (Top-K Entry)

*Let $J_K(s_t)$ be the baseline top-$K$ set at state $s_t$, and let $i \notin J_K(s_t)$. Define*

$$\mathcal{G}_{i,K} := \left\{\bar{h} \in \mathbb{R}^d : \#\left\{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\right\} \leq K - 1\right\}.$$

*If $\mathcal{G}_{i,K}$ contains a nonempty open set and the centered prompt law satisfies (S1), then*

$$\Pr_{\bar{h}}\left(\bar{h} \in \mathcal{G}_{i,K}\right) > 0.$$

*Proof.* Under the local affine surrogate, token $i$ belongs to the surrogate top-$K$ set if and only if at most $K - 1$ tokens outrank it, i.e., if and only if $\bar{h} \in \mathcal{G}_{i,K}$. For each subset $S$ of token indices with $i \notin S$ and $|S| \leq K - 1$, define

$$\mathcal{O}_S := \left\{\bar{h} : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h}) \text{ for all } j \in S,\ z_i^{\text{lin}}(\bar{h}) > z_j^{\text{lin}}(\bar{h}) \text{ for all } j \notin S \cup \{i\}\right\}.$$

Each $\mathcal{O}_S$ is an intersection of open half-spaces and is therefore open. Moreover,

$$\mathcal{G}_{i,K} = \bigcup_{S:\, i \notin S,\ |S| \leq K-1} \mathcal{O}_S,$$

so $\mathcal{G}_{i,K}$ is a finite union of open sets and is itself open. By assumption it contains a nonempty open set. Condition (S1) therefore implies

$$\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0,$$

which is exactly the event that token $i$ enters the surrogate top-$K$ set. □

### E.6.5 Proof of Corollary 2 (Sample Complexity)

*Let $\mu_i = \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$. Within $N$ independent RSP samples $\bar{h}_1, \ldots, \bar{h}_N$, the probability of at least one admission is at least $1 - (1 - \mu_i)^N \geq 1 - \exp(-N\mu_i)$.*

*Proof.* The samples $\{\bar{h}_n\}_{n=1}^N$ are independent, so

$$\Pr\left(\text{no admission within } N \text{ samples}\right) = \prod_{n=1}^{N} \Pr(\bar{h}_n \notin \mathcal{G}_{i,K}) = (1 - \mu_i)^N.$$

The complement gives the first inequality. The second inequality is standard: $1 - \mu_i \leq \exp(-\mu_i)$, hence $(1 - \mu_i)^N \leq \exp(-N\mu_i)$. □

### E.6.6 Trajectory-Level Implications

Since autoregressive generation composes conditional distributions across steps, perturbations at early steps can lead to distinct trajectories. Let $\mathcal{R}_{k,K}(x)$ denote the set of trajectories containing at least one token outside the baseline top-$K$ set. If RSP increases the accessibility of such tokens at early steps, then the probability of sampling trajectories in $\mathcal{R}_{k,K}(x)$ increases under repeated sampling. The key point is temporal compounding: one early top-$K$ entry changes the next decoding state, which changes the next candidate set, and so on. This is the formal bridge from local rank perturbations to global trajectory diversity.

# F Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 4 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

## F.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,h)}$ denote the post-softmax attention weight at layer $\ell$, head $h$, from query position $i$ to key position $j$, satisfying $\sum_j \alpha_{i,j}^{(\ell,h)} = 1$ under the causal mask. Let $Q$ and $R$ denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let $H$ be the number of heads. The per-token attention mass that query $i$ directs to each group at layer $\ell$ is

$$\mathrm{PTM}_Q[\ell, i] = \frac{1}{|Q|H} \sum_{h=1}^{H} \sum_{j \in Q} \alpha_{i,j}^{(\ell,h)}, \qquad \mathrm{PTM}_R[\ell, i] = \frac{1}{|R|H} \sum_{h=1}^{H} \sum_{j \in R} \alpha_{i,j}^{(\ell,h)}. \tag{9}$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

## F.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^{5}$ and $\{B_P^{(b)}\}_{b=1}^{5}$. For sample $s$ and target group $\bullet \in \{Q, R\}$, the value in cell $(b_L, b_P)$ is

$$M_\bullet^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \mathrm{PTM}_\bullet[\ell, i]. \tag{10}$$

Figure 4 reports the sample average of Eq. (10) over the 500 valid samples.

## F.3 Excluding the `\boxed{...}` Span

The reasoning position range terminates strictly before the final `\boxed{...}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{...}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

## F.4 Excluding the Final Two Layers

The layer axis further excludes the last two transformer layers. Two well-documented properties of late layers, none specific to our setup, motivate this choice.

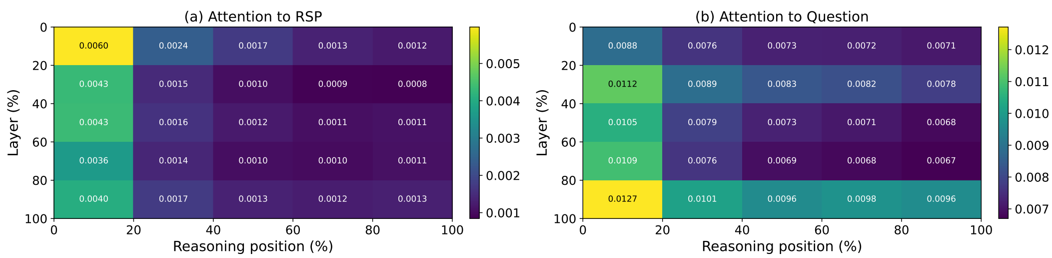
**Output-projection specialization.** Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

**Register-token hypothesis.** Uninformative patch tokens in Vision Transformers have been shown to be repurposed as *registers*: storage slots that absorb attention in deeper layers [Darcet et al., 2024]. Because RSPs share a similar profile as continuous tokens without pretrained semantic content, a similar repurposing in late transformer layers cannot be ruled out. Excluding the final two layers removes this potential register-induced confound from the reasoning-phase signal.

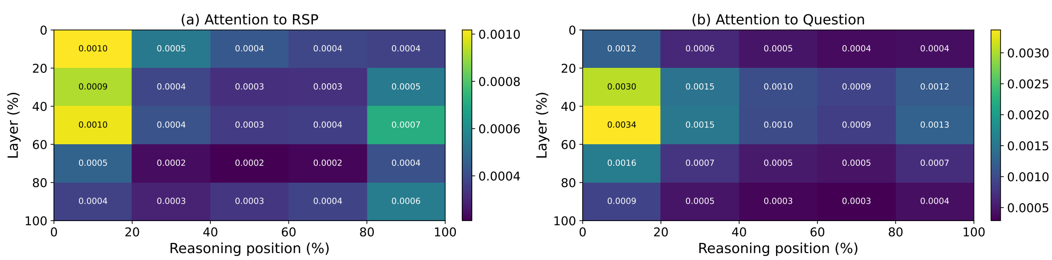
The two lines of evidence above identify known late-layer phenomena — linear readout and potential register repurposing — whose attention signatures are orthogonal to reasoning-phase dynamics. Removing the final two layers preempts their contamination of the reasoning-phase signal, yielding a principled, literature-grounded exclusion rather than a model-specific tuning decision.

### F.5 Additional Suffix Heatmaps

Figure 7 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across both models: question-token attention varies broadly across layers, whereas RSP attention shows the layer-wise and sequence-wise decay predicted by Eq. (3).




(a) Qwen2.5-Math-1.5B-Instruct, suffix




(b) LLaMA-3.1-8B-Instruct, suffix

Figure 7: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 4: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

## G GRPO Formulation

For a given prompt $q$, the behavior policy $\pi_{\theta_{\text{old}}}$ samples a group of $G$ responses $\{o_i\}_{i=1}^{G}$ with corresponding rewards $\{R_i\}_{i=1}^{G}$. The advantage of each response is computed by normalizing the group-level rewards:

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^{G})}{\text{std}(\{R_i\}_{i=1}^{G})}. \tag{11}$$

The policy is updated via a clipped objective with a KL penalty:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\Big(\min\Big(r_{i,t}\hat{A}_{i,t},\ \text{clip}(r_{i,t}, 1{-}\varepsilon, 1{+}\varepsilon)\,\hat{A}_{i,t}\Big) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})\Big)\right], \tag{12}$$

where $r_{i,t} = \pi_\theta(o_{i,t} \mid q, o_{i,<t})/\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$ is the importance sampling ratio, $\varepsilon$ is the clipping range, and $\beta$ controls the strength of the KL penalty against the reference policy $\pi_{\text{ref}}$.

## H Broader Impacts

This work provides empirical and theoretical analysis of how random embedding injection affects the reasoning behavior of large language models. The contributions are primarily foundational, with potential downstream implications for reinforcement learning-based post-training of reasoning models such as GRPO.

**Positive impacts.** RSP-induced trajectory diversity could improve the sample efficiency of RL-based LLM post-training by providing richer learning signals within group-relative advantage estimation. This has the potential to reduce compute costs associated with alignment and reasoning-oriented fine-tuning, making such methods more accessible to research groups with limited resources.

**Potential concerns.** Methods that expand the effective output support of LLMs may increase the reachability of low-probability tokens, which includes both desirable alternative reasoning paths and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems should combine it with existing safety filtering and alignment mechanisms rather than treating it as a standalone diversity source.

**Limitations on impact scope.** Since this work focuses on analysis rather than deploying a new model or dataset, the direct societal impact is limited. The broader implications described above are contingent on future work integrating RSP into applied training pipelines.

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